Theft, Trade, and Social Order*

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Abstract

What happens in a world without institutions that protect property rights? Hobbes predicted a war of all against all in which the incentive to produce withers. I model a production economy without external protection of property as a class of games in which individuals can trade or seize others' output by force. Familiar behaviors that govern a market emerge, even when the market is absent. In every subgame perfect equilibrium, individuals allocate productive labor as if jointly optimizing, equalize the return to labor across all goods, and gain nothing from isolation. The threat of theft extends the law of one price beyond markets: an implicit price of power is common to markets, conflicts, and isolated players.

What does not survive is the market's distribution of payoffs. In every equilibrium, each player's share of aggregate utility equals her power, her share of the economy's security, exactly as in universal conflict. Trade occurs only when market shares happen to coincide with power. Even under peaceful trade, every player arms as if she expected to fight. Disarmament and coordination within coalitions could make everyone better off, but some player can always do better still through force.

1 Introduction

Barry Weingast noted a fundamental political paradox: "A government strong enough to protect property rights and enforce contracts is also strong enough to confiscate the wealth of its citizens" (Weingast 1995). The paradox is not unique to government. In a hypothetical world without governments, and without other institutions that protect property, each individual must accrue power to protect her property. Power, once accrued, can be used either defensively or aggressively. Every individual who is strong enough to protect her property may be strong enough to confiscate the property of others. A government concentrates this dual use of power rather than resolving it.

Trade theory begins on the other side of this paradox, with property rights already secure. With secure property rights, the only natural alternative to trade in the classical formulation is withdrawal

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and isolation. The world we observe, however, contains both trade and conflict. Institutions that protect property constrain violence, but they have not eliminated it, and the dilemma of power persists wherever their reach ends.

How institutions came to constrain violence is difficult to study directly, because their origins predate the written record and may be permanently unrecoverable (Fukuyama 2011; Willhoite 1976).[1] This paper instead examines institutions by envisioning an economy without them. I retain the familiar production and exchange environment of neoclassical trade but remove the guarantee of secure property rights, allowing individuals to protect goods with security and to appropriate them through theft. In a production economy modeled as a game, players choose whether and how to interact with one another, generating conflicts, markets,[2] and isolation as equilibrium outcomes. The resulting economy provides a point of comparison for the institutions we observe.

Much of what we expect from a market economy survives in the "theft and trade" economy. Because specialization has the potential to improve outcomes for everyone, isolation gains nothing. Players make production choices as if they were cooperating, with no explicit coordination. They allocate labor to the options with the highest returns, which equalizes their individual valuations across all goods. The threat of theft gives the whole economy a common valuation, a law of one power price that holds in markets, conflicts, and isolation alike. The law of one price is its special case when applied to markets.

What changes in this economy is distribution. In every equilibrium, payoffs follow the proportionality principle of Cox et al. (2019): each player's payoff is in proportion to her power. Trade emerges only when the market allocation coincides with the power-weighted allocation, and nothing in a decentralized economy ensures that it does. Even when markets emerge, players arm as if they were preparing for conflict. I show that a schedule of transfers exists that can align the market allocation with the power-weighted allocation, satisfying the conditions for markets to exist in any theft and trade economy. Implementing the transfers could replace violence with trade, but coercion would not disappear. The consequences of power are preserved in the transfers themselves.

[1] Evidence for record-keeping and territorial markers extends into deep prehistory (Schmandt-Besserat 1978). Early inter-group and inter-species violence has left archaeological traces (Crevecoeur et al. 2021; Meijer 2024), though its causes and extent remain debated (Rutar 2023). Comparative evidence from non-human primates suggests the capacity to constrain intragroup violence may predate modern humans entirely (Masters 1991).

[2] A market is an institution in its own right. In this economy, it is an institution chosen by the players, not an external institution that limits their choices. A market is defined formally in Section 2.2.1.

Furthermore, the existence of this schedule establishes only the feasibility, not the implementation, of economy-wide peace.

Although the theft and trade economy allows players to organize and cooperate, individual incentives erode the gains that might have been achieved through cooperation. Coalitions that share goods equally among members encourage efficiency, but lack stability. Without external enforcement, the members of a coalition turn on each other. Coalitions that reward members by their contribution to the coalition's security eliminate the incentives to cooperate. Even the Pareto-efficient grand coalition falls to individual incentives. Any player can dominate the grand coalition with minimal cost. The double-edged nature of power reproduces Weingast's paradox in every configuration players can create on their own initiative.

The next subsection relates this paper's contributions to the literatures of conflict and institutions. Section 2 presents the model and formalizes the notion of theft. Section 3 establishes the properties of equilibria in the theft and trade economy, and Section 4 asks whether redistribution or organization can improve on the equilibrium outcome. The paper concludes in Section 5 with a discussion of the implications of conflict in an economy, and what an institution that sustains peace might accomplish.

1.1 Related Literature

Many fields, from paleoanthropology to archaeology to linguistics, supply evidence from which inferences about the origins of institutions are drawn. This paper contributes to the two literatures in economics and political economy that are most closely related to the emergence of property rights as an institution. The economic literature on conflict explores how individuals divide resources between producing goods and appropriating them. The institutional literature describes how existing and historical institutions constrain violence and sustain social order. This paper links them, using the tools of the conflict literature to construct an economy without external institutions, which provides a point of comparison for the institutional literature.

Skogh and Stuart (1982), Skaperdas (1992), De Meza and Gould (1992), Hirshleifer (1995), Grossman and Kim (1996), Grossman (2002), Bates et al. (2002), and Baker et al. (2010) all study

environments in which individuals must allocate scarce resources between producing goods and protecting or appropriating them. In these models, the threat of conflict defines the strategic environment in which allocation decisions are made. Hirshleifer, for example, explicitly excludes exchange to induce conflict over resources, and asks when the resulting anarchic order is sustainable. Piccione and Rubinstein (2007) take the opposite extreme where power alone determines who controls endowments, with no production, arming, or exchange.

Polachek (1980) established a literature in which nations choose between trade and conflict, which the later work of Martin et al. (2008) follows. A related literature examines the theft and trade choices of two parties, whether individuals, groups, or nations (Anderton, Anderton, et al. 1999; Anderton and Carter 2008; Hausken 2004; Rider 1999, 2002). Within the literature directly comparing trade and conflict, the closest antecedents to the present model are Skaperdas and Syropoulos (1996, 2001, 2002), who combine production, trade, and conflict to examine how insecure property rights affect the gains from trade and incentives to invest in security. They find that the prospect of trade can increase arming enough that agents prefer to forgo it. To examine which features of an economy rely on the choice between trade and competition over production, this paper constructs a general economy with an arbitrary finite number of agents and goods in which theft is always available. The generality reveals features of an economy that two-agent models cannot express.

The institutional literature has long connected political organization to the control of coercion and the distribution of economic resources. Max Weber (1946) described forms of political organization and canonically placed control over legitimate violence at the center of the modern state. Mancur Olson connected coercive capacity to both collective action and political organization: selective incentives help overcome the collective-action problem (Olson 1971), while the stationary-bandit metaphor illustrates how an actor capable of predation can acquire an incentive to establish order rather than continually appropriate output (Olson 1993; see also Allen et al. 2023; Sánchez de la Sierra 2020). Avner Greif examines the origins of various historical institutions capable of constraining violent behavior and supporting exchange (Greif 2006). Acemoglu and Robinson (2006) model voting, transfers, wealth distribution, revolutions, coups, and similar important institutions directly. Together, these authors explain how institutions constrain violence and sustain social order. This literature, however, generally takes the existence of such institutions as a starting

point.

North, Wallis, and Weingast, with additional contributors, place the control of violence at the center of their explanation of social order (North, Summerhill, et al. 2000; North, Wallis, Webb, et al. 2013; North, Wallis, and Weingast 2009; Wallis 2026). Individuals with the capacity for violence form organizations that control resources and threaten violence against others. Social order depends upon these organizations collecting rents so that powerful actors prefer cooperation to violence. This logic produces the proportionality principle (Cox et al. 2019). Stability requires that members of the dominant coalition receive rents in proportion to their relative power. Early versions of the principle appear in Skaperdas (1992) and Skaperdas and Syropoulos (1997) between two players dividing joint production. This paper extends proportionality to all actors and every form of interaction the players choose in a general economy. Trade occurs only when the market's division of output coincides with proportionality.

2 Model of Conflict and Trade

To develop modeling assumptions relevant to a world without externally enforced property rights, I turn to the intellectual tradition that explicitly considers such a world. Classical political philosophers debated the origins of justice, society, and individual rights; neoclassical economic models have generally taken the institutions that define and enforce those rights as given. Appendix A discusses how these debates shape the model's assumptions, and Appendix B collects the model's notation.

2.1 Game

The theft and trade model is a class of games that extends neoclassical trade (Ricardo 1888; Samuelson 1939) by incorporating theft as an option. A theft and trade game is a dynamic two-stage game with complete and imperfect information. Formally, the game

$$\Gamma = (\mathcal{N}, \Sigma, \pi) \tag{1}$$

is defined with the standard objects: the set of players, the strategy profile space and a payoff function mapping a strategy profile to a payoff profile. After a brief overview of the game, I explain each of these objects in detail.

In the first round, players produce M consumer goods, and one *security good*. A player's security, relative to the security of others, determines the likelihood she can successfully appropriate goods from other players or defend goods from appropriation. In the second round, players choose how they would like to interact with each other player. These round-one production and round-two interaction choices govern the allocation of goods among players, and players derive utility from their consumption.

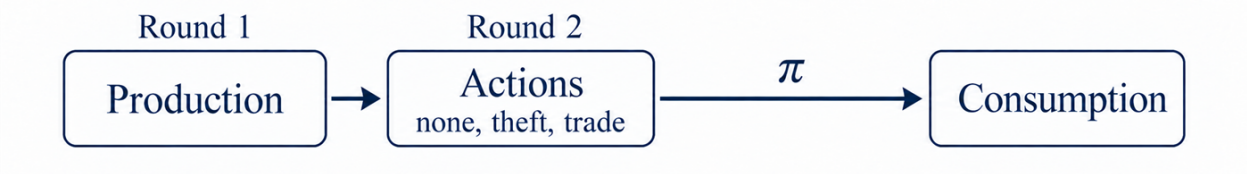


Figure 1: Game flow

2.1.1 Players

The set of players is $\mathcal{N} = \{1, 2, \ldots, N\}$, $N \geq 2$. Throughout, n denotes a generic player.

The players are self-interested and rational. They maximize expected utility. When participating in a market, players are price-takers, but otherwise players behave strategically[3].

2.1.2 Round One

In round one, each player simultaneously divides a fixed endowment of labor among the goods she can produce.

There are M types of fungible, transferable consumer goods in the economy, indexed by $m \in \mathcal{M} = \{1, 2, \ldots, M\}$, and one nontransferable *security good*. Players share a common utility function over consumer goods.

[3]Price-taking is one of the rules of the Market institution defined in Section 2.2.1, which players choose to join. The assumption maintains the comparison to the institution in neoclassical trade, so departures from classical results come from the availability of theft rather than from market power.

Assumption 1 ***Preferences*****:** *Utility $U : \mathbb{R}^M_+ \to \mathbb{R}_+$ is continuous, increasing, homogeneous of degree one, and normalized so $U(\mathbf{0}) = 0$; U is C^2 and strictly quasiconcave on $\mathbb{R}^M_{++}$. All consumer goods are essential: $U(x) > 0$ if and only if every component of x is positive.*

Nature endows each player with a fixed quantity of labor $\tau > 0$ and a production function for each good, $f_{nm} : \mathbb{R}_+ \to \mathbb{R}_+$ for each consumer good $m \in \mathcal{M}$ and $f_{nS} : \mathbb{R}_+ \to \mathbb{R}_+$ for the security good. These endowments are common knowledge.

Assumption 2 ***Technology*****:** *Each production function is continuous at 0, C^2 on $\mathbb{R}_{++}$, increasing, and strictly concave, with $f_{nm}(0) = f_{nS}(0) = 0$. Labor is the sole input; there are no intermediate or capital goods.*

Each player has an identical labor endowment τ[4] measured in units of time. In round one, players simultaneously allocate their labor endowments across production activities. Player n's labor allocation $L_n = (t_{n1}, \ldots, t_{nM}, s_n)$ assigns $t_{nm} \geq 0$ units of labor to good m and $s_n \geq 0$ units to security, exhausting her endowment:

$$s_n + \sum_{m \in \mathcal{M}} t_{nm} = \tau. \tag{2}$$

$\mathcal{L}_n$ denotes the set of player n's feasible allocations, and $\mathcal{L} = \prod_{i \in \mathcal{N}} \mathcal{L}_i$ denotes the set of labor allocation profiles $L = (L_1, \ldots, L_N)$.

These allocations cannot be revised once production begins. Production is complete before round two begins, at which point each player's labor allocation becomes common knowledge.

Player n's production bundle of consumer goods is $X^P_n = (f_{n1}(t_{n1}), \ldots, f_{nM}(t_{nM}))$, and her production of the security good, her *security*, is $S_n \equiv f_{nS}(s_n)$. The labor s_n that produces it is her *security budget*. For any set of players $I \subseteq \mathcal{N}$, $S_I \equiv \sum_{i \in I} S_i$ denotes their combined security.

2.1.3 Round Two

In round two, each player decides how to treat every other player.

[4] I impose identical labor endowments for simplicity and parsimony. Allowing heterogeneous labor endowments does not alter the substantive results.

Each player simultaneously chooses a *posture* toward each other player after observing every player's round-one labor allocation.[5] Player n's posture toward player $j \neq n$ is $a_{n,j} \in \{\text{none, theft, trade}\}$, and her action a_n is the $(N-1)$-vector of her postures. A contingent plan

$$\rho_n : \mathcal{L} \to \{\text{none, theft, trade}\}^{N-1}, \qquad a_n = \rho_n(L), \tag{3}$$

specifies player n's action at every labor allocation profile $L \in \mathcal{L}$. The players' actions together form the action profile $a = (a_1, \ldots, a_N)$.

2.1.4 Strategies and Strategy Profiles

Player n's strategy is an ordered pair consisting of her labor allocation in round one and her contingent plan for round two:

$$\sigma_n = (L_n, \rho_n) \in \Sigma_n, \qquad \sigma = (\sigma_1, \ldots, \sigma_N) \in \Sigma = \prod_{i \in \mathcal{N}} \Sigma_i. \tag{4}$$

I also write $\sigma = (L, \rho)$, where $L = (L_1, \ldots, L_N)$ and $\rho = (\rho_1, \ldots, \rho_N)$, with $\rho(L) = \big(\rho_1(L), \ldots, \rho_N(L)\big)$.

Every play of the game ends at a *terminal history* $h = (L, a) \in \mathcal{L} \times \mathcal{A}$, where $\mathcal{A} = \prod_{i \in \mathcal{N}} \{\text{none, theft, trade}\}^{N-1}$ is the set of action profiles. Each labor profile $L' \in \mathcal{L}$ begins a round-two subgame, and the contingent plans ρ end it at the terminal history $\big(L', \rho(L')\big)$. A strategy profile $\sigma = (L, \rho)$ therefore induces the terminal history $h(\sigma) = \big(L, \rho(L)\big)$ on its path, and specifies a terminal history in every subgame off it. Outcomes and payoffs are defined at terminal histories, and a strategy profile receives the payoffs of the terminal history it induces.

Players may choose any set of postures toward the other players, without restrictions requiring internal consistency. Imposing restrictions that ensure consistent choices in every situation would be cumbersome, and could make each player's feasible choices depend on the choices of others. A simpler approach is to impose consistency on the structure induced by the strategy profile rather than on the players' choices themselves. Imposing three consistency rules ensures that

[5] This choice is termed a posture because it is the interaction the player desires. However, the interaction between players that is realized depends on the postures of both players.

every terminal history induces a well-defined partition of players. The payoffs are then defined independently for each set of players in the partition. I explain these consistency rules in the following subsections as part of describing the partition.

2.2 Interaction Graph

The players' choices (along with the consistency rules) define the partition over the players that then defines the payoffs of the game. The language of graph theory is particularly convenient in describing the consistency rules and the partition the players' choices induce. The interaction graph below is a simple yet precise description of the partition, and not a necessary feature of the theft and trade model.

An action profile a induces an undirected interaction graph $G(a) = (\mathcal{N}, E^C(a), E^T(a))$ whose vertices are the players. The graph depends only on the action profile, not on the labor allocations. I write G, E^C, and E^T when the action profile is clear from context, and for a strategy profile $\sigma = (L, \rho)$, I write $G(\sigma) \equiv G(\rho(L))$, and likewise $E^C(\sigma)$ and $E^T(\sigma)$, for the graph on its path. If either player in a pair adopts a 'theft' posture toward the other, the pair is joined by a Conflict edge:

$$E^C = \{e_{ij} : a_{i,j} = \text{theft or } a_{j,i} = \text{theft},\ i, j \in \mathcal{N},\ i < j\}. \tag{5}$$

The behavioral assumptions underlying theft and trade make the two forms of interaction incompatible in this model. Trade requires mutual consent, and players who trade accept a property-rights arrangement in which each player owns her own production. Theft, in contrast, requires only unilateral action and cannot be refused. When a player chooses theft, she contests ownership of the goods produced by her counterpart. The outcome of theft determines who ultimately consumes the contested goods, but that outcome is not known when trade choices are implemented. While ownership remains contested, neither player can convey the uncontested property rights that trade requires. A player facing theft therefore cannot trade with another player, since she cannot establish ownership of the goods she would exchange. Consequently, a player joined to anyone by a Conflict edge cannot have any Trade edges. The first consistency rule is therefore that Conflict takes precedence over Trade. A pair is joined by a Trade edge only if both players adopt a 'trade'

posture toward each other and neither is joined to any player by a Conflict edge:

$$E^T = \{e_{ij} : a_{i,j} = a_{j,i} = \text{trade},\ e_{ik} \notin E^C \text{ and } e_{jk} \notin E^C\ \forall k \in \mathcal{N},\ i < j\}. \tag{6}$$

Players with no edges are Autarkic. Figure 2 illustrates an interaction graph.

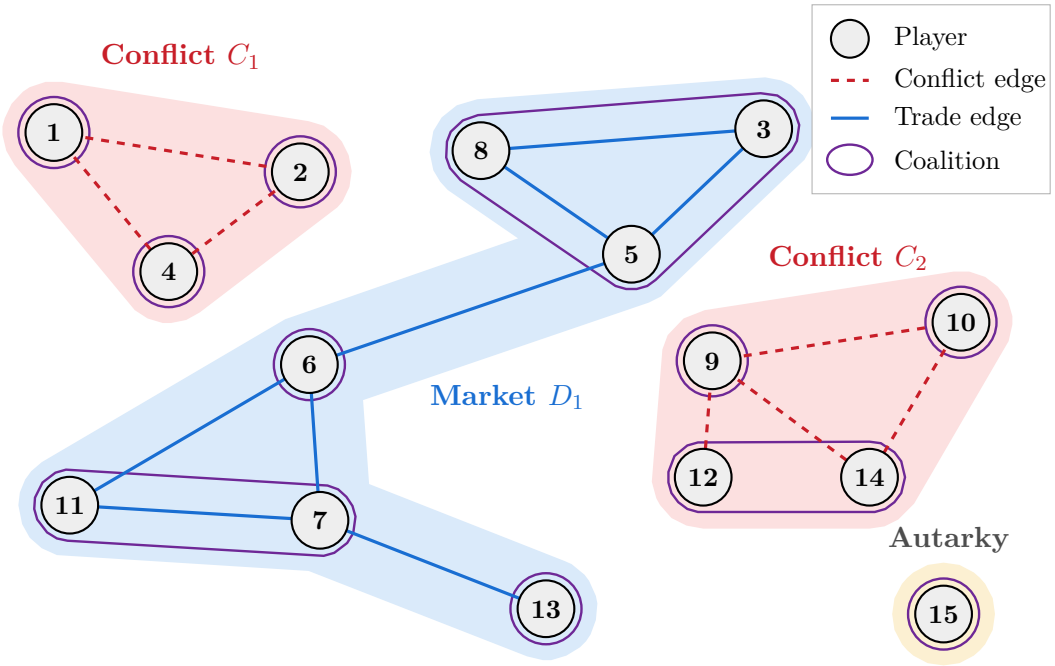


Figure 2: An interaction graph with two Conflicts, one Market, and one Autarkic player. Shaded regions are the connected components of G. Rings mark candidate coalitions: players 12 and 14 form a candidate within C_2, players 3, 5, and 8 and players 7 and 11 form candidates within D_1, and every other player is a singleton. No candidate contains a Conflict edge between its members.

2.2.1 Conflicts and Markets

The vertex sets of the connected components of the interaction graph form a partition of the players in the economy.

A *Conflict* C is a connected component of the subgraph $(\mathcal{N}, E^C)$ with at least two players, and a *Market* D is a connected component of the subgraph $(\mathcal{N}, E^T)$ with at least two players. Because no player can hold both kinds of edge, every connected component of G is either a Conflict, a Market, or a single Autarkic player. Let $\text{Comp}(G)$ denote the set of connected components of G. For any $H \in \text{Comp}(G)$, let $V(H)$ denote its vertex set. Thus, $V(C)$ is the set of combatants in Conflict C,

and $V(D)$ is the set of players trading in Market D.

Although the model creates an economy with no exogenous institutions that protect property rights, a Market is an institution. The Market institution accepts a particular form of property rights: players own their own production, and exchange requires bilateral consent. Players endogenously choose to participate in a Market institution defined by the following properties:

- Consumer goods are fungible within each type and freely transferable within a Market
- All players in a Market are price takers
- Each participant's spending cannot exceed the market value of their production
- Market clearing
- Exchange requires two-way consent

Players choose whether to abide by this version of property rights and are free to make a different choice. It is their own choice that establishes the right rather than an imposition by some external force or authority.

2.2.2 Coalition Structure

Within each component, players may act alone or as a group.

A coalition is a set of players who jointly optimize their labor allocations and postures and accept a rule for dividing their joint consumption allocation. A coalition therefore has three properties. The first two concern the structure of the players' interaction and can be checked at any terminal history $h = (L, a)$; the third concerns their behavior.

(i) Players accept the coalition's sharing rule; no member declares theft against another

(ii) Players must be in the same component of the interaction graph $G(a)$

(iii) At a strategy profile σ, members' labor allocations and contingent plans jointly maximize $\sum_{i \in B} \pi_i(\sigma)$, given the other players' strategies

Each game is indexed by a *coalition structure* $\mathcal{B}$, a partition of $\mathcal{N}$ whose elements are the *admissible* coalitions: the only sets of players the game permits to cooperate. Each admissible coalition B carries a sharing rule assigning each member $i \in B$ a share $\gamma_i \geq 0$ of the coalition's consumption allocation, with $\sum_{i \in B} \gamma_i = 1$.[6] I discuss γ further in the results, leaving it general for the moment. Although the admissible coalitions are fixed within each game, every subset of players is admissible in some game in the theft and trade class of games, so the model can examine the consequences of coalitions without specifying a process by which coalitions form.

Whether an admissible coalition can cooperate depends on how its members choose to interact. An admissible coalition $B \in \mathcal{B}$ is a *candidate coalition*, or simply a *candidate*, at the terminal history $h = (L, a)$ if it satisfies properties (i) and (ii), which together with Conflict precedence are the three consistency rules:

$$\exists\, H \in \mathrm{Comp}\left(G(a)\right) \text{ such that } B \subseteq V(H), \qquad \text{and} \qquad e_{ij} \notin E^C(a) \;\; \forall\, i \neq j \in B. \tag{7}$$

An admissible coalition that is not a candidate dissolves, and its members act as singletons. I treat each such singleton as a trivial candidate. The candidates then form a partition $\mathcal{B}(h)$ of $\mathcal{N}$ that refines the component partition of $G(a)$ and is defined at every terminal history, including those reached only off the equilibrium path. Property (iii) concerns strategies rather than terminal histories, and uses the payoff function π defined in the next section; it completes the definition of a coalition.

Definition 1 ***Coalition:*** *A candidate B with $|B| \geq 2$ on the path of a strategy profile σ is a* coalition *at σ if it satisfies property (iii).*

With no communication and no enforcement beyond the members' own power, nothing in a player's strategy declares that a candidate cooperates; a coalition can only be identified by its members' behavior. Because property (iii) includes members' contingent plans, their joint choice covers what they would do after every labor profile they could jointly choose, not only their postures on the equilibrium path. Each candidate shares its output by its sharing rule whether or not its members

[6] I treat the sharing rule as an institution in the sense of Greif (2006): no external actor enforces it, so it governs behavior only if it is self-enforcing, with every member preferring to accept it over declaring theft.

jointly optimize; a candidate whose members do not jointly optimize is not a coalition, but its sharing rule still determines their payoffs.[7] For a component H, $\mathcal{B}_H = \{B \in \mathcal{B}(h) : B \subseteq V(H)\}$ denotes the candidates within it. The rings in Figure 2 show one such partition.

Property (i) excludes admissible coalitions whose members contest one another's output. Members who steal from each other are not cooperating to achieve an outcome superior to what they can achieve individually; they are acting as individuals. Property (ii) excludes admissible coalitions that span components, and it follows from the same property-rights logic as Conflict precedence. A coalition pools its members' output, so its claim to the pool is no stronger than its claim to each member's contribution. If any member is in Conflict, the pool is contested: members trading in a Market cannot convey the uncontested property rights that trade requires, and an attack on a member in one Conflict is an attack on goods claimed by a member in another, merging the two contests. If no member is in Conflict, trading the pooled output requires the consent of every member, so a coalition cannot span two Markets, or a Market and an Autarkic player, whose members have not consented to trade with one another. The members of a candidate therefore share a common prize.

Because the partition $\mathcal{B}(h)$ contains both singletons and sets of players who share their output, the outcome function takes the candidate as the basic unit when assigning outcomes. Consumption bundles and payoffs, however, are ultimately assigned to individuals, which the sharing rule does. A singleton's share is one.

2.3 Payoffs

Payoffs are assigned in two steps. The outcome function g maps each terminal history to a lottery over consumption allocations, and the value function v maps each lottery to the players' expected utilities. A strategy profile is evaluated at the terminal history it induces, by the payoff function

$$\pi = v \circ g \circ h \tag{8}$$

[7] The results do not depend on the convention that the coalition dissolves completely. The deviations the analysis uses leave the deviating player a singleton, and under a ratio-form contest success function her probability of winning does not depend on how the other players are grouped.

In the subgame beginning at labor profile L', players receive $v(g(L', \rho(L')))$. Section 2.3.1 develops g; Section 2.3.2 defines v.

2.3.1 Outcome Function

The outcome function translates the goods produced by players into consumption allocations at each terminal history $h = (L, a)$. Throughout this subsection, fix h: production is that of the labor profile L, G denotes $G(a)$, and $\mathcal{B}_H$ is taken from $\mathcal{B}(h)$. Each vertex set $V(H)$ of a connected component H of G is a member of a partition over the set of players. Goods do not cross from one set in the partition to another. They are fully distributed among the players in the same partition set. Therefore, each lottery, g_H, is independent, and fully allocates aggregate production among the players in $V(H)$.

The operator $Y(\cdot)$ denotes the aggregate consumer good production of any set of players:

$$Y(I) \equiv \sum_{i \in I} X_i^P \qquad \text{for any } I \subseteq \mathcal{N}. \tag{9}$$

Although the operator $Y(\cdot)$ is defined for a set of players, there should be no ambiguity using a shortcut and allowing $Y(\cdot)$ to also operate on components H of G, $Y(H) \equiv Y(V(H))$. In a Market D, $Y(D)$ is the total supply of the Market. In a Conflict C, $Y(C)$ is the prize of the contest that the players $V(C)$ are competing for. For Autarkic player n, $Y(\{n\}) = X_n^P$ is simply n's consumer good production.

The outcome function g assigns a lottery g_H over candidates' consumption bundles to each connected component $H \in \text{Comp}(G)$. These component lotteries are independent, so the joint lottery over the economy-wide consumption allocation is their product:

$$g(h) = \bigotimes_{H \in \text{Comp}(G)} g_H(h). \tag{10}$$

Each candidate's bundle X_B^F is subdivided among its members by the sharing rule, so member n's final consumption bundle is

$$X_n^F = \gamma_n X_B^F. \tag{11}$$

An Autarkic player is an isolated vertex of G and therefore constitutes a singleton component whose lottery is degenerate at the player's production bundle: $X_n^F = X_n^P$.

Because players have common homothetic preferences, they have identical demand schedules, so any non-singleton coalition division — regardless of how γ is set — leaves each member's consumption bundle proportional to the candidate coalition's bundle X_B^F, simultaneously satisfying the demand schedule for each player.

The assumption of common homothetic preferences is sufficient to guarantee a unique price vector that clears a Market D (Chipman 1974; Gorman 1953; Kehoe 1985, p. 139; Quah 1993). Because candidates have identical demand schedules and markets clear, each candidate's consumption bundle is some proportion α_B of aggregate Market production $Y(D)$. The Market lottery is also degenerate, so for player $n \in B$,

$$X_n^F = \gamma_n \, \alpha_B \, Y(D). \tag{12}$$

For a Conflict C, the lottery is not degenerate. The contest is winner-take-all and non-destructive: the winning candidate receives the entire prize $Y(C)$, the aggregate production of consumer goods by all candidates in the Conflict[8]. The probability that candidate $B \in \mathcal{B}_C$ wins is determined by a ratio-form contest success function based on the candidates' security:

$$p_B = \begin{cases} \dfrac{S_B}{S_{V(C)}} & \text{if } S_{V(C)} > 0, \\ \dfrac{|B|}{|V(C)|} & \text{if } S_{V(C)} = 0. \end{cases} \tag{13}$$

The Conflict lottery g_C therefore awards the bundle $Y(C)$ to candidate B with probability p_B, for each $B \in \mathcal{B}_C$. A player's *power* is her probability of winning a Conflict that spans every player, $q_n \equiv S_n / S_{\mathcal{N}}$ (or $1/N$ if no player holds security), and $q_I \equiv \sum_{i \in I} q_i$ is the combined power of a set of players I. Power is relative: it depends on every player's security, not only her own.

[8] Any other division would require restraining the winner from taking the rest of the prize. Nothing in the model supplies that restraint. The contest is the last move of the game, so once it is decided, no player retains an action that could punish a winner who takes everything. A restraint would have to be imposed from outside, which is the kind of institution the model excludes.

2.3.2 Payoff Function

The value function v takes expectations over the outcome lottery: $v_n(g(h)) = E_{g(h)}\left[U(X_n^F)\right]$ is player n's expected utility of the lottery for the component she belongs to at the terminal history h. A strategy profile receives the payoffs of the terminal history it induces:

$$\pi_n(\sigma) = v_n\Big(g(h(\sigma))\Big) = E_{g(h(\sigma))}\left[U(X_n^F)\right] \qquad \forall\, n \in \mathcal{N}, \tag{14}$$

and $\pi = (\pi_1, \ldots, \pi_N) : \Sigma \to \mathbb{R}_+^N$ is the payoff function of the game Γ. In the subgame beginning at any labor profile L', the players receive $v\Big(g(L', \rho(L'))\Big)$. Payoffs depend on the coalition structure and sharing rules through the outcome function; I write $\pi_n(\sigma)$ and suppress this dependence when it is clear from context.

2.4 Equilibrium Concept

I analyze pure-strategy subgame perfect equilibria of the theft and trade game. I examine coalition stability against unilateral deviations without establishing coalition-proofness.

For each coalition structure $\mathcal{B}$ and sharing rule γ, payoffs are defined at every terminal history, so each pair $(\mathcal{B}, \gamma)$ indexes a game $\Gamma(\mathcal{B}, \gamma)$ in the theft and trade class. A strategy profile σ^* is a subgame perfect equilibrium of the game $\Gamma(\mathcal{B}, \gamma)$ if its restriction to every subgame is a Nash equilibrium of that subgame (Osborne and Rubinstein 1994)[9].

A *Trade equilibrium* is an SPE where all players trade.

Definition 2 ***Trade Equilibrium*:** *A Trade Equilibrium is a subgame perfect equilibrium in which* $E^C(\sigma^*) = \emptyset$

A *Social Order equilibrium*, on the other hand, is an SPE where some players trade. A Social Order equilibrium allows the coexistence of Trade and Conflict.

Definition 3 ***Social Order Equilibrium*:** *A Social Order equilibrium is a subgame perfect equilibrium in which* $E^T(\sigma^*) \neq \emptyset$

[9] Unqualified, *equilibrium* means a subgame perfect equilibrium.

Therefore, a Trade equilibrium is a Social Order equilibrium where no Conflict is present. Neither definition excludes Autarkic players; whether they arise is determined by equilibrium behavior.

2.5 Remarks

Before turning to equilibrium analysis, there are a few implications of the model's assumptions that should be explicit.

Remark 1 ***Positive Security*****:** *In equilibrium, each player who is a singleton allocates a positive amount of labor to security, $s_n > 0$. A player choosing to forgo security while working alone invites aggression: any other player with a positive amount of security wins a bilateral conflict with certainty and takes both players' production. Forgoing security therefore leaves the unprotected player with zero utility, whereas allocating even a small positive amount of labor to security yields a positive expected payoff regardless of the interaction.*

Remark 2 ***No Market for Security*****:** *There is no market for security goods. Under Conflict, players do not exchange goods. Once engaged in Trade, a player has no remaining use for security, and the labor devoted to its production is a sunk resource cost with no consumption benefit. No market participant would therefore surrender consumer goods in exchange for security. Because Trade and Conflict are mutually exclusive outcomes of the same stage, no market for security forms.*

Remark 3 ***Risk Attitude*****:** *I make no assumption about risk attitudes, however, the utility assumptions imply that players behave as if risk-neutral over the lotteries that arise. Players maximize expected utility with a utility function that is concave (Lemma C.2 (Concavity of U)) and homogeneous of degree one, so their attitude toward risk depends on which lotteries are compared. For a general lottery that gives bundle x with probability λ and bundle y otherwise, concavity gives $\lambda U(x) + (1-\lambda)U(y) \leq U\big(\lambda x + (1-\lambda)y\big)$. A player strictly prefers the expected bundle to the lottery unless x and y are proportional. When x and y are proportional, homogeneity makes U linear along a ray from the origin and the player is indifferent. Outcomes in the model are a joint lottery composed of independent component lotteries, and each player's payoff depends only on the sublottery of the component she belongs to. Each component lottery is either degenerate or awards*

a pooled bundle in full, so the bundles in each component lottery lie on a single ray from the origin. Along these lotteries players behave as if risk neutral.

3 Equilibrium Analysis

3.1 Equilibrium Characterization

The most noticeable consequence of replacing Autarky with Conflict is a fundamental change in who receives the gains. In classical markets, preferences and productive capacities jointly determine the allocation of goods through the price mechanism. In the theft and trade economy, power determines payoffs. By declaring theft against everyone, any player can unilaterally turn the entire economy into the war of all against all that Hobbes called Warre, regardless of the strategies adopted by others. Whatever she could secure in that single Conflict is therefore a lower bound on what she will accept in any SPE. As the following theorem demonstrates, these individual payoff constraints characterize the payoffs of every SPE.

Formally, let

$$\pi_n^C(L) = q_n(s)\, U\big(Y(\mathcal{N})\big)$$

denote player n's payoff when every player belongs to a single Conflict at labor profile $L \in \mathcal{L}$, where $q_n(s)$ is player n's power at security profile s. The *universal-Conflict labor-allocation game* is the game in which each player n chooses $L_n \in \mathcal{L}_n$ and receives $\pi_n^C(L)$.

Theorem 1 ***Characterization:*** *In an arbitrary theft and trade game with N players and M consumer goods, any SPE $\sigma^* = (\sigma_1^*, \ldots, \sigma_N^*)$ satisfies*

$$\pi_n(\sigma^*) = q_n(s^*)U(Y(\mathcal{N})) \qquad \forall\, n \in \mathcal{N},$$

where s^* is the security allocation profile at a Nash equilibrium L^* of the universal-Conflict labor-allocation game.

A characterization of every SPE would be empty if no SPE existed. Standard existence results require payoffs that change continuously with players' choices, but the payoff functions of the

theft and trade game are discontinuous at $s = \mathbf{0}$. A player with no security loses everything to anyone holding even a little. I therefore prove the existence of an SPE directly, by constructing an equilibrium characterized by universal Conflict—Hobbesian Warre.

Proposition 2 ***Existence of Warre SPE*****:** *There exists a strategy profile $\sigma^* = (L^*, \rho^*)$ satisfying the following conditions:*

(i) ρ^* is common to every player and, for every $L \in \mathcal{L}$ and every $n, j \in \mathcal{N}$ with $n \neq j$,

$$a_{n,j} = \textit{theft}$$

(ii) L^* is a Nash equilibrium of the universal-Conflict labor-allocation game: for every $i \in \mathcal{N}$,

$$\pi_i^C(L_i^*, L_{-i}^*) \geq \pi_i^C(L_i, L_{-i}^*) \qquad \forall\, L_i \in \mathcal{L}_i.$$

Every such strategy profile is a subgame perfect equilibrium of the theft and trade game.

Because every pair of players is joined by a Conflict edge at every labor profile, every admissible coalition with two or more members dissolves, and the Warre profile is a subgame perfect equilibrium for any coalition structure and sharing rule.

In a universal Conflict, players allocate labor to balance the size of the contest prize against their expected shares of that prize. Theorem 1 (Characterization) shows that this balance governs every SPE. Each player's expected share of aggregate utility is exactly her power, her probability of winning a universal Conflict, as in Warre. This distribution of payoffs holds in Conflict, Trade, or Autarky. Theorem 1 extends power-weighted payoffs from the Warre equilibrium to SPEs that induce arbitrary partitions of $\mathcal{N}$ into Markets, Conflicts, and Autarkic singletons. Every subgame perfect equilibrium of a theft and trade game reproduces the labor allocation in addition to the power-weighted payoffs of Hobbesian Warre. Neither Trade nor Autarky offers an escape from this distributional constraint. Every SPE is therefore payoff-equivalent to the *power-weighted allocation*, in which each player receives the bundle $q_n\, Y(\mathcal{N})$[10].

[10] By homogeneity, $U\big(q_n Y(\mathcal{N})\big) = q_n\, U\big(Y(\mathcal{N})\big)$.

The power-weighted allocation is where the distinction between Autarky and Conflict as outside options is felt. Autarky permits a player to disengage completely, but at the cost of forgoing the gains generated by specialization. The loss incurred from withdrawing is only beneficial in carefully constructed cases. Conflict is an outside option that binds. In a theft and trade economy, a player can claim a share of aggregate production that is determined by her power rather than by her productive input to aggregate production.

Cox et al. (2019) name this allocation of surplus by power the *proportionality principle* and identify it as a necessary condition among the elite organizations that constitute the dominant coalition. The earlier versions noted in Section 1.1 (Skaperdas 1992; Skaperdas and Syropoulos 1997) arise in economies in which agents individually produce arms but jointly produce a single consumption good, so that the common product is divided according to relative arms. In the theft and trade economy, agents independently produce both security and multiple consumption goods, and Autarky remains feasible. Theorem 1 extends the proportionality principle beyond elite organizations. The principle arises from the outside options of individual agents before organizations exist and, as Section 4 will show, continues to apply when those agents organize. The power-weighted allocation is not just a condition governing the distribution of rents among powerful elites, but an equilibrium restriction on payoffs wherever the threat of theft remains unconstrained.

The proof of Theorem 1 (Characterization) rests on an identity: payoffs aggregate within arbitrary connected components, which allows the theorem to accommodate any equilibrium partition of the players.

Lemma 1 ***Aggregate Payoff Identity*****:** *For any terminal history $h = (L, a)$ and any connected component H of $G(a)$,*

$$\sum_{n \in V(H)} v_n(g(h)) = U(Y(H)).$$

In particular, for any strategy profile σ and any component H of $G(\sigma)$, $\sum_{n \in V(H)} \pi_n(\sigma) = U(Y(H))$.

The identity says that, once production is fixed, the form of interaction is purely distributive. A Market and a Conflict consisting of the same players realize the same aggregate value $U(Y(H))$, and only the division differs. Differences in total value can therefore arise only from individual production choices and the choice of which players interact with one another.

3.2 Economy Properties

Combining Theorem 1 (Characterization) with Lemma 1 (Aggregate Payoff Identity) reveals a common valuation of security across the economy, even when its participants do not trade. The implicit *power price* is the expected utility a player realizes per unit of security. For any component H, its power price is the utility of its aggregate production divided by its members' aggregate security. At an arbitrary strategy profile, power prices might differ across components, and a player's own payoff per unit of security can differ from other players in the same component. In equilibrium, however, this valuation must equalize. Security is never traded, so no Market sets this valuation, but power governs payoffs the way prices govern allocations.

Corollary 1 ***Law of One Power Price*****:** *Let σ^* be an SPE, and for any component H of $G(\sigma^*)$ let $q_H \equiv \sum_{n \in V(H)} q_n(s^*)$ denote its members' combined power. Then*

$$Y(H) = q_H \, Y(\mathcal{N}) \qquad \text{and} \qquad U\big(Y(H)\big) = q_H \, U\big(Y(\mathcal{N})\big).$$

Every component's consumer-good production is the same fraction of aggregate production as its members' combined power, and its members realize exactly that share of aggregate utility.

Because any player can revert to Warre, the law of one power price holds across every component in the economy: Markets, Conflicts, and isolation. In equilibrium, the members of each component produce the same proportion of consumption goods relative to aggregate production in the entire economy as they produce security goods relative to security good production in the entire economy. Each component's output therefore reflects the composition of the entire economy's output.

The law of one power price then restricts actual prices in Markets in the theft and trade economy.

Corollary 2 ***Law of One Price*****:** *In any SPE, all Markets clear at the same relative prices. Moreover, the common price vector is proportional to $\nabla U\big(Y(\mathcal{N})\big)$.*

Consider the classical market argument. If different prices emerge in different markets, there is an opportunity to profit via arbitrage. The arbitrageur moves goods between disconnected markets until all prices equalize in both markets. Arbitrage is crippled in the theft and trade economy,

however. If players had the bilateral consent necessary to trade across markets, the players would already have trade links, i.e. the market would not be fragmented, and arbitrage would not be necessary. A unilateral deviation to establish trade is not sufficient because the would-be arbitrageur needs a willing trade partner in a disconnected market to establish the trade link.

Off the equilibrium path, if the power price varies across components, players have a profitable opportunity. A player in a component with a low power price can declare theft against the rest of the economy and claim her share of the larger prize. This deviation works the way arbitrage operates in a Market economy. Arbitrage profits from a difference in prices between markets, and theft profits from a difference in the power price between components; in equilibrium, neither difference can survive. In the theft and trade economy the power price, rather than arbitrage, disciplines prices in the market.

Because Markets describe the entire classical trade economy, the classical law of one price begins and ends with Markets. When Conflict enters the picture, an economy can fragment along multiple fault lines. Because the threat of theft holds whether people are peacefully trading, isolating themselves, or already in conflict, the law of one power price establishes a common valuation across the economy as a whole, and the law of one price is its special case where prices are established through exchange.

The law of one power price places a particularly strong restriction on Autarky. The universal requirement that every player's equilibrium payoff be power-weighted eliminates Autarky as an equilibrium choice except when a player has no gains from specialization left to exploit. Any Autarkic player in a proposed equilibrium must receive a payoff determined by her power and aggregate production, rather than by her own production alone. If gains from specialization remain across her labor allocation, she can unilaterally reallocate labor to increase aggregate utility without altering anyone's power, then initiate Conflict to capture her share of the increase.

Corollary 3 ***Autarky Exclusion*****:** *If an SPE contains an Autarkic player n, then for every pair of consumption goods m, m',*

$$\frac{f'_{nm}(t_{nm})}{f'_{nm'}(t_{nm'})} = \frac{\partial U(Y(\mathcal{N}))/\partial X_{m'}}{\partial U(Y(\mathcal{N}))/\partial X_m}.$$

Thus, Autarky can occur in equilibrium only if no gains from unilateral specialization remain for that player.

In a Market economy, an isolated producer stays isolated only if her own rates of transformation already match the economy's prices, so that specializing and trading would gain her nothing. Corollary 3 (Autarky Exclusion) shows that an isolated player's marginal rates of transformation must equal the marginal rates of substitution of the common utility at aggregate production, the same condition a Market economy with representative agents imposes through prices. Any gap between the two is a gain from specialization: shifting her labor toward the goods she produces relatively cheaply would raise the value of aggregate production. But the force driving the equality differs. A classical producer is drawn out of isolation by the mutual gains from trade[11]. Here, any gains from specialization an isolated player leaves unexploited are gains she can claim by reallocating her labor and declaring theft, the omnipresent option. In equilibrium, the law of one power price therefore requires every component, including a player who stands alone, to produce goods in the same composition as aggregate production. Isolation can be an equilibrium choice only when a player's own optimal production already has that composition, leaving no gains from specialization to claim.

One of the central ideas of economics is that specialization generates surplus that can make everyone better off. In a Market, the invisible hand leads self-interested traders to allocate labor as if they were jointly maximizing aggregate production, because exchange benefits all parties. Conflict offers no such mutual gain. A contest has a winner and a loser, and every unit one combatant takes is a unit another loses. If the invisible hand works through mutual gain, it is reasonable to expect Conflict to block it. The next proposition shows that no such obstacle exists. Holding security fixed, combatants allocate labor across consumer goods exactly as traders do.

Proposition 3 ***Common Labor Allocation*****:** *Fix a security allocation s with $0 < s_i \leq \tau$ for every $i \in \mathcal{N}$ and $s_i < \tau$ for at least one i. Consider the production games induced by a single connected Market and a single universal Conflict among all players in $\mathcal{N}$. The two games have the same*

[11] In the neoclassical trade economy, someone else is always willing to trade.

unique Nash equilibrium in consumption-good labor allocations:

$$t^{T*} = t^{C*}.$$

A Market allocates players' pooled production through the price mechanism; a Conflict awards the same pooled bundle in a lottery with security-weighted probabilities. In a Market, prices equal marginal utilities at the aggregate bundle, so a player who maximizes the value of her own output also maximizes her contribution to aggregate utility. In a Conflict, a player's expected share of the prize is fixed by security. Once security is chosen, the only thing her production choice can change is the value of the prize. Trade and Conflict differ in how the pie is split, but players in either continuation want to maximize the value of the pie.

Proposition 3 (Common Labor Allocation) holds security fixed. Even though security has no counterpart in classical trade, players allocate labor to it by a familiar principle.

Proposition 4 ***Security Budget*****:** *Let L^* be a Nash equilibrium of the universal-Conflict labor-allocation game. Every player holds positive security, $s_n^* > 0$, and for every consumer good m,*

$$1 - q_n(s^*) \;\geq\; \frac{\partial \ln U(Y(\mathcal{N}))/\partial t_{nm}}{d \ln f_{nS}(s_n^*)/ds_n}, \qquad \textit{with equality if } t_{nm}^* > 0,$$

where derivatives at $t_{nm} = 0$ are right derivatives.

At the optimum, the cost of security equals its value, with both measured in percentage terms. Its cost is set by her technology: the elasticity of the prize with respect to her security along her production frontier, the percentage of the prize she forgoes for each additional percent of security. Its value is set by the contest: her chance of winning is her share of total security, and her own security counts toward that total, so a one percent increase in her security raises her chance of winning by $1 - q_n$ percent, her probability of losing the universal Conflict. Because that value does not depend on which good the labor is drawn from, security costs her the same proportion of the prize in every good she produces, and her labor earns the same return in every use. This is the counterpart of a competitive producer setting her marginal rate of transformation equal to relative prices: technology determines the rate at which she can trade one use of labor for another, and the

contest, rather than a Market, determines the rate at which that trade is rewarded.[12]

The characterization holds for every player, whatever regime she ultimately occupies. Security is chosen in round one, before players determine who trades and who fights. Security is not tradeable or consumable and therefore matters only because of the Conflicts a player could be drawn into. When every interaction can be turned into Conflict, even a player who ends up trading holds the security she would want if she expected to fight. Security is a deterrent, required whether or not the Conflict materializes.

3.3 Social Order

The power-weighted allocation generated by the threat of theft establishes familiar properties of a Market economy in a new setting. The invisible hand guides players to coordinated production, prices agree across Markets, a common concept of value permeates the economy, and isolation gains nothing. Whether players actually trade is a separate question. Given security, a Market produces exactly the bundle a Conflict among the same players would (Proposition 3), and every player's payoff is fixed by her power (Theorem 1). A Market therefore has no more surplus to offer its members than Conflict does. Since the pie is the same size, any member who receives more in the Market than her power-weighted share does so at another member's expense, and the member who loses would rather steal than accept the loss. A Market emerges only if the market allocation, each member's share α_n of the Market's output at market-clearing prices (12), happens to match the power-weighted allocation.

Proposition 5 ***Conditions for Social Order:*** *A Social Order equilibrium exists if and only if there exist a Nash equilibrium L^* of the universal-Conflict labor-allocation game and a set $I \subseteq \mathcal{N}$ with $|I| \geq 2$ such that, at L^*,*

(i) $Y(I) = \left(\sum_{i \in I} q_i(s^*)\right) Y(\mathcal{N})$, *and*

(ii) when the players of I form a Market, $\alpha_i = \dfrac{q_i(s^*)}{\sum_{j \in I} q_j(s^*)}$ *for every $i \in I$.*

[12] When marginal products at zero labor are finite, a player may devote all of her labor to security; Appendix D characterizes the corners where this occurs.

Corollary 4 ***Conditions for Trade*****:** *A Trade equilibrium exists if and only if, at some Nash equilibrium L^* of the universal-Conflict labor-allocation game, $\alpha_i = q_i(s^*)$ for every $i \in \mathcal{N}$ when all players form a single Market.*

Proposition 5 (Conditions for Social Order) and Corollary 4 (Conditions for Trade) turn the necessary condition of Theorem 1 (Characterization) into exact characterizations of when Markets can exist. The two conditions required for Social Order operate at different levels. Condition (i) is a condition between groups. At equilibrium, every component is a scaled version of the economy. The output, the security, and the utility its members realize are each the same fraction of the economy as a whole (Corollary 1 (Law of One Power Price)). For players who wish to trade, this is condition (i) for social order. Condition (ii) makes trade fragile. Within the Market, each member's share of its output must also match her power. When the entire economy forms a single Market, the between-group condition is satisfied, leaving only the requirement that each player's market share equal her power for the Trade equilibrium to emerge. Markets can form in units smaller than the entire economy, but only when the Market reproduces both a scaled version of economy-wide production and the power-weighted allocation.

Sufficiency rests on a simple property of common homothetic preferences—no group of players values its own output more than the market does. A player who steals from her trading partners expects a power-weighted share of their joint production, and because specialized trading partners produce lopsided bundles, that share is worth no more to her than its market value. When each trader's market share already equals her power, seizing her partners' output gains her nothing that trade does not already provide. Specialization, the source of the gains from trade, also makes theft from a subset of trading partners unattractive. Stealing from too few members of a Market yields a lopsided bundle worth less than the value it could have commanded in the broader market, so targeting too small a subset of the market reduces the thief's expected reward. Stealing from the entire Market yields an expected share of a bundle with the economy's proportions, so even theft at full scale only breaks even.

Both conditions of Proposition 5 (Conditions for Social Order) are equalities and nothing in the primitives of the model requires either to hold. Market shares depend on the value of each player's productive contribution, while power depends on security technologies and the opportunity cost

of diverting labor from consumption-good production. A player whose labor produces particularly valuable output may earn a large market share, but also faces a high opportunity cost of allocating that labor to security. Productive advantage and coercive advantage need not coincide. Social Order is therefore possible but fragile: it requires an alignment between market shares and power that the economy has no mechanism to ensure.

The equilibrium properties of the theft and trade economy bring together two intellectual traditions that are normally seen as incompatible. Hobbes imagined life without institutions as "solitary, poor, nasty, brutish, and short" (Hobbes 2017, ch. 13). In Hobbesian *Warre*, even those who are not inclined toward violence must be prepared to fight for survival. The ever-present threat of conflict undermines incentives to produce. Classical economics offers a different account of self-interest, in which individually optimizing behavior generates outcomes resembling cooperation. Specialization and exchange create gains from trade, isolation is costly for everyone, and arbitrage generates benefits even among individuals who never interact directly.

In the theft and trade economy, many of the benefits of trade survive in the Hobbesian setting where the threat or exercise of violence is inescapable. Autarky is unappealing because specialization serves individual self-interest. Players make production choices as if they were coordinating. The law of one power price regulates prices across fragmented markets and beyond. The threat of theft disciplines the entire economy, not just its markets. Whenever payoffs diverge from power, someone has an incentive to exploit the difference.

What does not survive is the market determination of payoffs. Preferences continue to determine relative prices, and those prices continue to guide production choices. But payoffs are governed by power rather than by the market value of each player's productive contribution. Whether peaceful exchange can be sustained depends on whether the allocation it delivers matches what players could obtain through theft. A purely self-interested player might forgo the chance of seizing the entire prize through violence, but only when peaceful exchange delivers the full value of her expected payoff from theft.

Introducing Conflict as a choice changes payoffs in a way that is neither classical nor Hobbesian. Power-weighted payoffs emerge from self-interested choices regardless of how players ultimately interact. Each player weighs the larger expected share that more security buys against the consumer

goods she gives up to produce the larger share. Although Conflict destroys nothing directly, fighting is risky, as a combatant stakes everything she produces on the outcome of the contest and the losers are left with nothing. Trade avoids that gamble but not the opportunity cost of Conflict. Security is chosen against the threat of Conflict rather than its occurrence, so a player who trades holds exactly the security she would hold if she expected to fight, and the labor diverted from production is lost either way. Peace is therefore no cheaper than war.

4 Redistribution and Organization

Theorem 1 (Characterization) fixes payoffs by power, and Proposition 5 (Conditions for Social Order) shows that the market delivers the power-weighted allocation only by coincidence. Two possibilities might nevertheless sustain trade and alleviate the deadweight loss of arming. First, redistribution could bring the market allocation into alignment with the distribution dictated by power, eliminating the incentive to appropriate. Second, coordination of production and security choices could create a larger prize, potentially leaving room for peaceful cooperation.

Both possibilities require an arrangement that can withstand the players' individual incentives, but the only enforcement technology available in the model is security itself. This section shows why neither possibility escapes that constraint. A redistribution to the power-weighted allocation always exists, but the model provides no mechanism to implement it. Coordination creates genuine potential gains, but only a power-weighted division of the surplus can survive individual incentives. Even that division reproduces the security expenditures of Hobbesian *Warre*. The opportunity to appropriate one's power-weighted share prevents players from sustaining the gains that coordination could otherwise achieve.

4.1 Redistribution

The market is one way of dividing what the economy produces. Theorem 1 (Characterization) identifies the division that equilibrium requires, and nothing prevents the economy's output from being divided that way directly.

Corollary 5 ***Transfer Schedule*****:** *For any strategy profile σ, there exists a transfer schedule of consumer goods, applied to the final allocation at σ, under which every player's payoff equals $q_n U(Y(\mathcal{N}))$.*

Whatever the players produce, and however they interact, the economy's output can be divided as Theorem 1 (Characterization) requires. The obstacle to exchange is a disagreement between rewarding economic productivity and appeasing power. Such a transfer schedule would remove the disagreement by reproducing the power-weighted allocation. A schedule keyed to power, however, only reproduces the division coercion would deliver anyway.

Redistribution takes the economy's output as given. Organization asks whether it can be made larger. In Markets and Conflicts alike, players allocate consumer-good labor as if they were jointly optimizing (Proposition 3 (Common Labor Allocation)). Given security, each player's share of the prize is fixed, so her only interest is the size of the prize. Security is different. It changes shares rather than the prize, so each player's security budget choice affects the payoff of the others. Extending optimization to security budget choices could create gains from specialization and increase surplus in the economy.

Because specialization is required to achieve this additional surplus, one might hope for a division that rewards productivity, or divides surplus more equitably. Any such division must be held in place against players who would do better by force. However, a coalition's sharing rule is an institution its members accept like the Market. The coalition must survive its own members' choices.

4.2 Shares within Coalitions

Proposition 6 ***Shares Must Match Power*****:** *Suppose a coalition B, with $|B| \geq 2$, forms on the path of a strategy profile σ. If $\gamma_i \neq S_i/S_B$ for any $i \in B$, then σ is not a subgame perfect equilibrium.*

Every player, whether in a coalition or acting alone, can turn her component into a Conflict by declaring theft against everyone in it, and doing so guarantees her a share of the component's

output proportional to her power. Although a jointly-optimizing coalition increases the surplus of the entire market, the members' power-weighted claims exhaust the output - including any additional surplus. So no member can be paid less than her power commands, and no payoff can exceed the power-weighting without another falling below. Under equal shares, for example, a member whose share of the coalition's security exceeds her share of its headcount does better fighting on her own, where her win probability reflects her power rather than her presence. Within a Market, a member who attacks only her partners forfeits the coalition's gains from trade with the rest of the Market, which might deter a narrowly targeted defection. However, a coalition member who attacks the whole Market forfeits nothing, and the guaranteed power-weighted share of the entire market applies. Whatever rule a coalition adopts, it cannot hold members' shares anywhere other than where power puts them. The result extends the payoff characterization of Theorem 1 to the internal organization of coalitions: even when players work together, their shares must reflect coercive power.

Power-weighted sharing removes one obstacle to coalition stability. Can such coalitions generate additional surplus?

Lemma 2 ***Coalition Optimization*****:** *Fix a Conflict C and a candidate $B \subseteq V(C)$ whose sharing rule assigns each member a share $\gamma_i > 0$ that does not depend on the labor allocation. Hold the action profile, and with it the interaction graph and the partition into candidates, and every labor allocation outside B fixed, and suppose $S_{-B} \equiv \sum_{j \in V(C) \setminus B} S_j > 0$. A profile of labor allocations $(L_i)_{i \in B}$ is a Nash equilibrium of the induced game among the members of B if and only if it maximizes the candidate's joint payoff $p_B\, U\big(Y(C)\big)$ over $\prod_{i \in B} \mathcal{L}_i$.*

Fixed shares align individual incentives with the coalition objective. Every member's payoff is a positive multiple of the same objective, so the coalition is a team in the sense of Marschak and Radner (1972), and Proposition 6 (Shares Must Match Power) applies the result that person-by-person optimality implies team optimality under concavity (Radner 1962). Because labor supply is fixed, every unit of a member's labor goes to something the coalition values, and no member has a private use of her time on which to free-ride. Free-riding cannot explain any failure of organization in this model. But efficiency is not the only requirement a coalition must meet. Each member makes her own choice between cooperation and appropriation, and a coalition that does not survive those

choices cannot be a coalition in equilibrium.

Together, Proposition 6 (Shares Must Match Power) and Lemma 2 (Coalition Optimization) expose the problem with coalition formation. The only sharing rule consistent with coalition stability eliminates the gains from specializing in security. Under fixed shares, coalition members do not lose productivity to free-riding. Under power-weighted shares, however, members allocate labor to improve their share of the common prize rather than maximize its size. The resulting rent-seeking dissipates the additional surplus that the coalition could have produced.

4.3 Coalition Stability

The coalition with the most to gain from coordination is the whole economy. With no outside rival, it has no use for security at all.

Proposition 7 ***Grand Coalition Instability*****:** *If $\mathcal{N}$ is a coalition at a strategy profile σ, then σ is not a subgame perfect equilibrium.*

The grand coalition's joint plan reaches the efficient frontier by disarming, and disarmament invites conquest. Against an unarmed economy, a player who holds an arbitrarily small amount of security and declares theft against everyone else wins the entire economy's output with certainty. The efficient frontier is reachable only by a coalition that cannot defend it. The efficiency of the Classical trade result is out of reach. Every theft and trade economy endures the loss from arming.

Proposition 8 ***No Productive Coalitions*****:** *In any SPE, every member of every coalition receives the payoff, and chooses the labor allocation, she would as a lone combatant. Consequently, no SPE contains a coalition consisting of two or more members, with at least one member who produces a consumer good.*

The logic is visible in the shares themselves. Once a coalition's shares match its members' power, each member receives exactly what she would earn alone: her share of the coalition's prize, $\frac{S_i}{S_B} \cdot \frac{S_B}{S_\mathcal{N}} U(Y(\mathcal{N}))$, telescopes to $\frac{S_i}{S_\mathcal{N}} U(Y(\mathcal{N}))$. Membership then changes nothing about her incentives, so she optimizes as if alone. But optimizing alone is not optimizing jointly. Each member's private return to security includes the share she captures from her partners, a transfer that nets to zero

across the coalition, so members invest more in security than the coalition's joint plan would. This is the collective rent dissipation of Nitzan (1991), generalized by Skaperdas (1996) and Münster (2009). A coalition that survives its members' incentives to defect cannot also be one that jointly optimizes. The one exception, a coalition of players who produce only security, accomplishes nothing and is discussed in Appendix D. The gains that organization would create are not hypothetical, and they would benefit every player. However, organization is not achievable within a theft and trade economy.

Redistribution and organization fail together. Corollary 5 (Transfer Schedule) can divide any prize, including the efficient one, but the size of the prize is settled in round one, before any division takes place. Security must be chosen against the possibility that it will be used, so every player arms whether or not she ever fights, and the gains from peace are lost before anyone decides whether to fight. Making disarmament safe requires more than redistributing output. Disarmament requires making each player's payoff independent of her own power, which would require a constraint on Conflict itself. The only enforcement technology the players possess is their own security, but every voluntary arrangement of players in the economy reproduces the power-weighted payoffs, eliminating the gains that might have been achieved through cooperation.

5 Discussion and Conclusion

This paper begins with a deceptively simple premise for an economic environment: introduce theft as an interaction option. Players produce consumption goods or security, and they keep, exchange, or appropriate what they produce. Let them interact in a universe where the only restraints on their choices are the ones they construct themselves. The formal machinery to deliver on this premise is complex. The purpose of this construction is not to describe an economy without rules, but to ask which rules emerge when rational and self-interested players choose their interactions only once. Because restraint is not imposed from outside, each feature of the equilibrium can be traced to the players' own incentives, and each feature of observed economies that the model does not produce points to something the model leaves out.

5.1 What the Model Answers

The model gives clear and somewhat unexpected answers to several questions. The first is how much of the market economy depends on secure property. Much less than one might expect. Whether players Trade or fight, their production choices make them behave as if they were jointly optimizing the value of aggregate production. Specialization continues to generate gains, the law of one price survives in Markets, and the law of one power price extends a common valuation to Conflict and isolation. Arbitrage disappears in this economy, but the threat of theft fills the role arbitrage plays in a Classical economy. Should a profitable subset of the economy emerge, a self-interested player would recognize the opportunity and take an action that disperses the excess profits.

The second is what secure property does provide. The answer is distribution. Autarky never challenges trade as a choice. Because Conflict has a bite, the Market distribution is challenged when players have the opportunity to choose between them. Every subgame perfect equilibrium is payoff-equivalent to the power-weighted allocation, and the proportionality principle applies across the entire economy. Markets form only when the market allocation matches the expectation of Conflict, and players who trade arm as if they expected to fight. A schedule of transfers consistent with universal peace exists, but nothing in the model implements transfers. One can imagine an enforcer who could reallocate market rewards to follow the power-weighted allocation and achieving peace, but if the enforcer requires power to implement the transfers, then a self-interested enforcer becomes another recipient in the power-weighted allocation.

The third is whether players can organize their way to a different outcome. Within the model, they cannot. Coalitions create an opportunity for cooperation and additional specialization, but each member's incentive to arm against her own partners dissipates the gains that might have been achieved. The grand coalition cannot disarm without inviting conquest.

5.2 Welfare

Two welfare implications follow from the equilibrium results in a theft and trade game. The first is that the equilibrium allocation is efficient once security is chosen. Given security, players allocate consumer-good labor as if they were jointly maximizing $U\left(Y(\mathcal{N})\right)$ (Proposition 3 (Common

Labor Allocation)), and in every SPE the resulting payoffs sum to $U(Y(\mathcal{N}))$ (Lemma 1 (Aggregate Payoff Identity), Corollary 1 (Law of One Power Price)). Because utility is common, concave, and homogeneous of degree one, no division of a bundle yields more total utility than the bundle itself (Lemma C.5 (Superadditivity of U)). Holding security fixed, then, no reallocation of consumer-good labor or of output can make one player better off without making another worse off. This holds in every SPE, whether players trade or fight. It also holds after Conflicts are resolved, not only in expectation, because every component's output is proportional to aggregate output, so the bundle each Conflict awards is too. In a Market, this is the familiar efficiency of competitive equilibrium. In a Conflict, the common value of security does the work that prices do in a Market.

The second implication is that the economy's inefficiency lies entirely in security. Every SPE is Pareto-dominated by disarmament under the same division of output and the loss from arming does not depend on how players interact. In any SPE, the labor allocation is a Nash equilibrium of the universal-Conflict game, so a player who trades holds the security she would hold if she expected to fight, and at a given labor allocation Trade and Warre deliver the same payoffs. Peace yields no dividend. Conflict destroys nothing, and the gains from specialization are realized whether players trade or fight, so the only resource peace could recover is the labor devoted to security. A peaceful equilibrium does not recover any labor, however, because the security that labor produces is necessary to achieve the peaceful outcome. Any reduction in security from a peaceful equilibrium would entice appropriation.

In this model, the cost of insecure property is the cost of arming, not the cost of fighting. A mechanism that replaced Conflict with Trade at the same division of output would not, by itself, improve welfare. A mechanism that reduced security would, and reducing security requires payoffs that do not depend on each player's own power. The welfare implications of theft are inextricably tied to the distributional implications.

5.3 Power

The model contains exactly one mechanism capable of securing property against theft: security. Adding security changes the power-weighting that equilibrium must respect, but does not separate payoffs from power. Assigning security to an enforcer does not resolve the problem when the

enforcer's members retain the same coercive opportunities and incentives as those they constrain. Their capacity to protect property remains a capacity to appropriate it. Weingast's dilemma thus arises without a state. The model cannot establish that no other mechanism could sustain a different allocation. It does indicate what such a mechanism would have to accomplish: sustain a distribution of goods that is not determined solely by the coercive power used to enforce it.

The comparison with observed economies requires care. Classical trade theory takes endowments and perfectly enforced property rights as given, leaving the mechanism that secures those rights unexplained. The theft and trade model makes property contestable, but allows only security to protect it. Neither framework explains the institutional arrangements under which theft and trade coexist in observed societies. Nor is the competitive allocation a benchmark independent of power: the property rights from which trade begins may themselves reflect whatever power holds them in place. The relevant question is whether the power needed to enforce an allocation must also determine the shares within it. In the theft and trade model, it must. In observed economies, the relationship is less clear.

Weingast formulated his dilemma from within an Open Access Order, where institutions protect property while constraining the exercise of coercion. Such societies appear to sustain distributions that do not simply reproduce each participant's coercive capacity, although establishing that departure would require a measure of power and the payoffs it commands. If that appearance is correct, some mechanism sustains an arrangement that security alone cannot sustain within the model. Yet the institutions that impose these constraints must themselves survive the coercive incentives that make them necessary.

Observed societies offer a partial counterpart to the transfer schedule of Corollary 5 (Transfer Schedule). They routinely pay specialists in violence: a military to deter external threats and police to limit theft and violence within. But these payments do not constitute the complete schedule. Institutions compensate designated enforcers, rather than every participant according to her coercive capacity. The model draws no such distinction. Security serves defense and appropriation alike, and the schedule assigns every player the bundle her power would claim, however she might use it. Within the model, partial transfers cannot sustain a departure from power-weighted payoffs: wherever a player receives less than her power commands, appropriation remains available (Theo-

rem 1 (Characterization)). Observed institutions nevertheless sustain an order in which property receives protection, unauthorized theft is punished but persists, and transfers compensate only some holders of power. The complete transfer schedule therefore does not explain how they sustain that order.

The same problem remains among the specialists institutions do pay. Compensation alone does not explain why enforcers accept limits on what they can claim. A military strong enough to protect property may also be strong enough to confiscate it, and a coalition within the model cannot sustain shares that differ from its members' relative security (Proposition 6 (Shares Must Match Power)). Designating some holders of power as legitimate enforcers supplies no additional restraint within the model. Observed institutions draw this distinction and appear capable of sustaining it, constraining both those authorized to use force and those excluded from that authority. How they do so remains unexplained. The model identifies the obstacle: whatever sustains these arrangements must limit the claims that coercive power can make, including the claims of those entrusted with enforcement.

5.4 Conclusion

The theft and trade model isolates a limiting case rather than reproducing an observed economy. Its narrow condition for Trade and its exact distributional restriction identify two questions for further work: how the costs of conflict widen the set of sustainable allocations, and how arrangements that constrain appropriation survive the incentives of those who enforce them.

The distributional restriction rests on a knife-edge. Players behave as if risk-neutral over the lotteries that arise (Remark 3 (Risk Attitude)), and Conflict destroys nothing. Labor diverted to security is its only resource cost. Destruction or risk aversion would make Conflict worth less than the power-weighted share of aggregate production that it delivers here, relaxing the payoff restriction and potentially admitting a range of divisions consistent with peaceful exchange. The mechanisms differ. Destruction reduces the contested prize and changes the return to security; risk aversion makes a sure allocation more valuable than an actuarially equivalent gamble. Neither extension establishes that competitive Markets would fall within the resulting range, or that lower security expenditures would follow. Those are questions for the extended model. The power-weighted allocation provides the benchmark as the additional costs of conflict vanish.

A broader question concerns the mechanisms that sustain exchange and organization. Players here choose once. Repeated interaction could make future exchange valuable enough to deter current appropriation, although it would also allow coercive power to change over time. Their preferences are fixed and material. Norms and legitimacy could alter the value they attach to appropriation or to respecting a claim. Coalitions pool individual production technologies. Cooperation that enables production unavailable to individuals acting alone could give members something valuable to lose by defecting. These mechanisms may also explain how institutions work: by changing the consequences of appropriation, the value of restraint, or the productive opportunities that cooperation preserves. Each possibility must confront the same problem, including among those entrusted with enforcement.

The theft and trade model shows that specialization, coordinated production choices, and common valuations arise from exchange - even when the exchange is involuntary. The model exposes the difficulty in escaping from a coercive distribution of goods when individuals are both self-interested and powerful. Observed institutions appear capable of establishing an imperfect version of property rights where both theft and trade coexist in a social order. But institutions are produced by self-interested individuals and wield the power that enables appropriation. Somehow institutions are capable of escaping the power-weighted allocation without succumbing entirely to rent seeking or collapse. The theft and trade economy shows how much economic coordination can emerge without institutions protecting property rights, and why concentrating coercive power among equally self-interested actors does not, by itself, free an economy from the distribution imposed by power. What institutions contribute lies in the mechanisms that overcome this constraint.

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A Classical Political Theory

I begin by assuming that individuals are self-interested and rational: they pursue their own ends and choose actions they believe will best achieve them. These are basic economic assumptions, but they are grounded in the philosophy of Machiavelli, Hobbes, and Hume. Machiavelli especially analyzes human action as revealed by what people do rather than by what society thinks they should do (Machiavelli 2003). Hobbes sets aside social pressures and assumes that individuals are motivated by their own desires (Hobbes 2017). Hume separates ends from means: people pursue their desires and employ their higher faculties to better acquire the things they want rather than to determine what it is they should want (Hume 2016). In this sense, self-interest determines the objectives and rationality governs the methods.

While Locke might agree with Hobbes that "the voluntary actions, and inclinations of all men, tend, not only to the procuring, but also to the assuring of a contented life," (Hobbes 2017), Locke constrains the pursuit of those ends by natural law. Individuals have a natural entitlement to their own labor and, through that labor, may acquire property in what they produce (Locke 2018). Grotius reaches a similar conclusion with an important qualification: production establishes ownership only when the other inputs to production do not already belong to another (Grotius 2024). Thus, for both Locke and Grotius, individuals may acquire and defend a property claim through their own productive activity even in the absence of institutions backing that claim.

The inevitability of conflict in Hobbes's *Leviathan* arises because some people will not respect that claim to a property right without a central authority. When one can never be certain who might attempt theft, Hobbes asserts that people "anticipate" that theft and act with preemptive violence. Hume similarly argues that without the institutions protecting property, individuals "wou'd run head-long into every kind of injustice and violence" (Hume 2016, T 3.2.2.21). Their solutions differ, but both identify the insecurity of possession as a source of conflict. Both Locke and Grotius require consent for the legitimate transfer of an individual's possessions; taking those possessions without consent provides just grounds for violence in response. It is that reactive violence that constrains theft in Locke's natural state without institutions. Any encroachment on one's natural rights could be met with escalated violence, for "men who judge their own case will always [. . .] overestimate

the amount of harm they have suffered" (Nozick 1974). The injured party may gather others to enforce the natural law, bringing sufficient power to bear against the transgressor.

The market institution defined in Section 2.2.1 embeds a particular form of property rights: participants recognize claims to the goods they produce and acquire through consensual exchange, and transfers require mutual consent. Players choose whether to respect these claims through their actions. Trade respects ownership by requiring consent; theft disregards the owner's claim by attempting appropriation without consent. The model therefore specifies the property rules governing market exchange without assuming that all players respect them.

The power individuals bring to conflict depends on their capabilities. Hobbes, like Rousseau, recognizes innate differences in those capabilities. Rousseau explicitly assumes inequalities "established by nature," including differences in health, physical strength, and qualities of mind (Rousseau 2004). Unlike Hobbes, Locke and Hume, Rousseau does not require these natural differences to be small. Hobbes claims that "the weakest has strength enough to kill the strongest"(Hobbes 2017, chapter 13, "Of the Natural Condition of Mankind."). I synthesize those viewpoints; although individual capabilities are not constrained in my model, conflict is treated as a stochastic contest; differences in capability affect the probability of success rather than determine its outcome.

Machiavelli counsels his prince to train himself and his armies in the arts of war through study and action, even more while at peace than during a war. Individual capabilities can be improved through effort. The capacity for violence is an endogenous choice, but not a costless one. Montesquieu suggests that commerce and military power pull against each other: "Peace is the natural effect of trade" (Montesquieu 2001, bk. 20, ch. 2). I incorporate these insights into my model. Individuals can produce 'security' through labor, but that labor is constrained through a limited endowment. Any labor spent developing security cannot be used to produce the things the individual desires.

While individuals can produce each of the goods they desire, producing all of those goods individually is not efficient. Plato begins his account of the origin of society from precisely this observation: "no one is self-sufficing, but all of us have many wants" (Plato 2000, bk. 2, 369b). Because individuals have different needs and different productive abilities, specialization allows them to satisfy those needs through the production of goods for others. I incorporate the same insight by assuming that individuals value a variety of goods. This creates potential gains from specialization and

exchange and, consequently, a peaceful alternative to acquiring the production of others through appropriation. The model asks which alternative individuals choose when both appropriation and exchange are available actions.

B Notation

Symbol	Meaning
Players, goods, and production	
$\mathcal{N} = \{1, \ldots, N\}$	Set of players; n a generic player, i, j, k indices
$\mathcal{M} = \{1, \ldots, M\}$	Set of consumer goods, indexed by m
U	Common utility function over consumer goods (Assumption 1)
τ	Labor endowment of each player
f_{nm}, f_{nS}	Player n's production functions for good m and for security (Assumption 2)
$L_n = (t_{n1}, \ldots, t_{nM}, s_n)$	Player n's labor allocation; $\mathcal{L}_n$ feasible set, $\mathcal{L}$ profile space
X_n^P	Player n's production bundle of consumer goods
$S_n = f_{nS}(s_n)$	Player n's security; s_n her security budget; $S_I = \sum_{i \in I} S_i$
Strategies	
$a_{n,j}$, a_n, a	Posture toward j; action; action profile
ρ_n	Contingent plan mapping labor profiles to actions
$\sigma_n = (L_n, \rho_n)$, σ	Strategy; strategy profile in $\Sigma = \prod_i \Sigma_i$
Interaction structure	
$h = (L, a)$, $h(\sigma)$	Terminal history; terminal history on the path of σ
$G(a) = (\mathcal{N}, E^C, E^T)$	Interaction graph with Conflict and Trade edges; $G(\sigma) \equiv G(\rho(L))$
$\mathrm{Comp}(G)$, $V(H)$	Connected components of G; players in component H
C, D	A Conflict; a Market
$\mathcal{B}$	Coalition structure: the admissible coalitions
$\mathcal{B}(h)$, $\mathcal{B}_H$	Candidates at h; candidates within component H
$Y(I)$	Aggregate consumer-good production of set I; $Y(H) \equiv Y(V(H))$
Outcomes and payoffs	
γ_i	Member i's share under an admissible coalition's sharing rule
α_B	Candidate B's share of Market production
p_B	Candidate B's probability of winning its Conflict
q_n	Player n's power, S_n / S_M; $q_I = \sum_{i \in I} q_i$
g, g_H	Outcome function; lottery for component H
$u_n(g(h))$	Player n's expected utility of the outcome lottery at terminal history h

C Proofs

C.1 Proofs for Section 3

Several short lemmas establish properties of the CSF and Utility functions that are useful in other results.

Lemma C.1 ***Concavity of the Contest Share*****:** *Fix a Conflict C, a candidate $B \subseteq V(C)$, and the labor allocations of every player outside B, with $S_{-B} \equiv \sum_{j \in V(C) \setminus B} S_j > 0$. The contest share*

$$p_B = \frac{S_B}{S_B + S_{-B}}, \qquad S_B = \sum_{i \in B} f_{iS}(s_i),$$

is strictly concave in the members' security $(s_i)_{i \in B}$. In particular, for a singleton $B = \{n\}$, p_n is strictly concave in s_n.

Proof. Let $s \neq s'$ be two profiles of the members' security and $\lambda \in (0,1)$. Each f_{iS} is concave, and strictly concave for every i with $s_i \neq s'_i$, of which there is at least one, so

$$S_B(\lambda s + (1-\lambda)s') > \lambda S_B(s) + (1-\lambda) S_B(s').$$

Let $g(x) = x/(x + S_{-B})$ for $x \geq 0$. Because $S_{-B} > 0$, $g'(x) = S_{-B}/(x + S_{-B})^2 > 0$ and $g''(x) = -2S_{-B}/(x + S_{-B})^3 < 0$, so g is strictly increasing and concave. Since $p_B = g(S_B)$,

$$p_B(\lambda s + (1-\lambda)s') > g(\lambda S_B(s) + (1-\lambda) S_B(s')) \geq \lambda\, p_B(s) + (1-\lambda)\, p_B(s'),$$

where the first inequality uses that g is strictly increasing and the second that g is concave.

■

Lemma C.2 ***Concavity of*** U**:** *U is concave.*

Proof. Concavity of utility under these assumptions is a standard result (Friedman 1973; Prada-Sarmiento 2011). The proof is included for completeness. Fix x, y in the domain of U and $\lambda \in (0,1)$.

First, suppose $U(x) = 0$. Since $x \geq \mathbf{0}$, $y + \frac{\lambda}{1-\lambda}x \geq y$ componentwise, so $U\Big(y + \frac{\lambda}{1-\lambda}x\Big) \geq U(y)$ by monotonicity. Homogeneity of degree one gives

$$U(\lambda x + (1-\lambda)y) = (1-\lambda)\,U\Big(y + \tfrac{\lambda}{1-\lambda}x\Big) \geq (1-\lambda)U(y) = \lambda U(x) + (1-\lambda)U(y).$$

The case $U(y) = 0$ is symmetric.

Now consider $U(x), U(y) > 0$. Since U is homogeneous of degree one, $U(x/U(x)) = U(y/U(y)) = 1$. By quasiconcavity of U, for every $\mu \in (0,1)$,

$$U\Big(\mu\,\frac{x}{U(x)} + (1-\mu)\,\frac{y}{U(y)}\Big) \geq 1.$$

Taking $\mu = \dfrac{\lambda U(x)}{\lambda U(x) + (1-\lambda)U(y)} \in (0,1)$ and applying homogeneity of degree one to clear the denominator gives

$$U(\lambda x + (1-\lambda)y) \geq \lambda U(x) + (1-\lambda)U(y).$$

■

Lemma C.3 ***Concavity of Utility in Labor*****:** *For any set of players $I \subseteq \mathcal{N}$ and any $H \supseteq I$, holding fixed the labor of players outside I, $U\big(Y(H)\big)$ is concave in the consumer-good labor of the players in I.*

Proof. Each component of $Y(H)$ is a sum of concave production functions of the players' consumer-good labor plus a constant, and is therefore concave. Let t and t' be two consumer-good labor profiles of the players in I and $\lambda \in (0,1)$. Componentwise, $Y(H)\big(\lambda t + (1-\lambda)t'\big) \geq \lambda Y(H)(t) + (1-\lambda)Y(H)(t')$, so since U is nondecreasing and concave (Lemma C.2 (Concavity of U)),

$$U\Big(Y(H)\big(\lambda t + (1-\lambda)t'\big)\Big) \geq U\Big(\lambda Y(H)(t) + (1-\lambda)Y(H)(t')\Big) \geq \lambda U\big(Y(H)(t)\big) + (1-\lambda)U\big(Y(H)(t')\big).$$

■

Lemma C.4 ***Positive Marginal Utility*****:** *U is strictly increasing in each good on $\mathbb{R}^M_{++}$, and at any $x \in \mathbb{R}^M_{++}$, $\frac{\partial U(x)}{\partial x_m} > 0$ for every $m \in \mathcal{M}$.*

Proof. Fix $x \in \mathbb{R}^M_{++}$ and let e_m be the mth unit vector. Suppose $U(x + c\,e_m) = U(x)$ for some $c > 0$. Since U is increasing, U is constant on the segment between x and $x + c\,e_m$,

which lies in $\mathbb{R}^M_{++}$; this contradicts strict quasiconcavity at the segment's midpoint. Hence U is strictly increasing in each good on $\mathbb{R}^M_{++}$.

Now suppose $\frac{\partial U(x)}{\partial x_m} = 0$. Since U is concave (Lemma C.2 (Concavity of U)) and differentiable at x, it lies below its tangent plane there, so

$$U(x + e_m) \leq U(x) + \frac{\partial U(x)}{\partial x_m} = U(x),$$

contradicting strict monotonicity.

■

Lemma C.5 ***Superadditivity of*** U**:** *For any partition of* $\mathcal{N}$ *into sets* H, $\sum_H U(Y(H)) \leq U(Y(\mathcal{N}))$.

Proof. For x, y in the domain of U, concavity (Lemma C.2 (Concavity of U)) and homogeneity of degree one give $U(x + y) = 2\,U\left(\frac{1}{2}x + \frac{1}{2}y\right) \geq U(x) + U(y)$. Applying this repeatedly to the sets H, whose $Y(H)$ sum to $Y(\mathcal{N})$, gives the claim.

■

Lemma 1 ***Aggregate Payoff Identity*****:** *For any terminal history* $h = (L, a)$ *and any connected component* H *of* $G(a)$,

$$\sum_{n \in V(H)} v_n(g(h)) = U(Y(H)).$$

In particular, for any strategy profile σ *and any component* H *of* $G(\sigma)$, $\sum_{n \in V(H)} \pi_n(\sigma) = U(Y(H))$.

Proof. Fix any terminal history $h = (L, a)$ and let H be an arbitrary connected component of $G(a)$. Let $\mathcal{B}_H$ denote the partition of $V(H)$ into candidates. For any player $n \in B \in \mathcal{B}_H$,

$$v_n(g(h)) = \begin{cases} \gamma_n \alpha_B U(Y(H)) & \text{if } H \text{ is a Market,} \\ \gamma_n p_B U(Y(H)) & \text{if } H \text{ is a Conflict,} \\ U(Y(H)) & \text{if } H = \{n\} \text{ is Autarkic.} \end{cases}$$

By construction, $\sum_{n \in B} \gamma_n = 1$ within each candidate, while candidate shares sum to one within each component: $\sum_{B \in \mathcal{B}_H} \alpha_B = 1$ in Markets and $\sum_{B \in \mathcal{B}_H} p_B = 1$ in Conflicts. Summing first over players within candidates and then over candidates therefore gives

$$\sum_{n \in V(H)} v_n(g(h)) = U(Y(H)).$$

The identity holds trivially for Autarkic players.

■

Theorem 1 ***Characterization:*** *In an arbitrary theft and trade game with N players and M consumer goods, any SPE $\sigma^* = (\sigma_1^*, \ldots, \sigma_N^*)$ satisfies*

$$\pi_n(\sigma^*) = q_n(s^*)U(Y(\mathcal{N})) \qquad \forall\, n \in \mathcal{N},$$

where s^* is the security allocation profile at a Nash equilibrium L^* of the universal-Conflict labor-allocation game.

Proof of Theorem 1 (Characterization). Let $\sigma^* = (\sigma_1^*, \ldots, \sigma_N^*)$ be an SPE in a theft and trade game. Fix $n \in \mathcal{N}$. At the second-round history following L^*, player n can unilaterally deviate to

$$a'_{n,j} = \text{theft} \qquad \forall\, j \neq n.$$

This deviation induces a single Conflict spanning $\mathcal{N}$, regardless of the other players' actions. Since no candidate contains players connected by a Conflict edge, player n is necessarily a singleton. With production already complete, the deviation therefore yields expected payoff $q_n(s^*)U(Y(\mathcal{N}))$.

Since σ^* is an SPE, n's equilibrium payoff must be no worse than this available deviation:

$$\pi_n(\sigma^*) \geq q_n(s^*)U(Y(\mathcal{N})) \qquad \forall\, n \in \mathcal{N}. \tag{15}$$

By superadditivity (Lemma C.5 (Superadditivity of U)),

$$\sum_H U(Y(H)) \leq U(Y(\mathcal{N})).$$

By Lemma 1 (Aggregate Payoff Identity), summing over the connected components H of $G(\sigma^*)$, which partition $\mathcal{N}$,

$$\sum_{n \in \mathcal{N}} \pi_n(\sigma^*) = \sum_H U(Y(H)).$$

Summing (15) over all players and using $\sum_{n \in \mathcal{N}} q_n(s^*) = 1$ gives

$$U(Y(\mathcal{N})) \leq \sum_{n \in \mathcal{N}} \pi_n(\sigma^*) \leq U(Y(\mathcal{N})).$$

Both inequalities must therefore bind. Since (15) holds for every player, equality of the sums forces equality in every term:

$$\pi_n(\sigma^*) = q_n(s^*)U(Y(\mathcal{N})) \qquad \forall\, n \in \mathcal{N}.$$

Now consider any feasible first-round deviation $\widehat{L}_n \in \mathcal{L}_n$. Player n can combine this labor allocation with a contingent action rule that declares theft toward every other player. Regardless of the other players' actions, this deviation yields

$$q_n(\widehat{s}_n, s^*_{-n})U\left(X_n^P(\widehat{L}_n) + \sum_{i \neq n} X_i^P(L_i^*)\right).$$

Since σ^* is an SPE, this deviation cannot be profitable. Substituting the payoff identity established above,

$$q_n(s^*)U\left(\sum_{i \in \mathcal{N}} X_i^P(L_i^*)\right) \geq q_n(\widehat{s}_n, s^*_{-n})U\left(X_n^P(\widehat{L}_n) + \sum_{i \neq n} X_i^P(L_i^*)\right) \qquad \forall\, \widehat{L}_n \in \mathcal{L}_n.$$

Thus L^* is a Nash equilibrium of the universal-Conflict labor-allocation game.

■

Proposition 2 ***Existence of Warre SPE*:** *There exists a strategy profile $\sigma^* = (L^*, \rho^*)$ satisfying the following conditions:*

(i) ρ^* is common to every player and, for every $L \in \mathcal{L}$ and every $n, j \in \mathcal{N}$ with $n \neq j$,

$$a_{n,j} = \textit{theft}$$

(ii) L^* is a Nash equilibrium of the universal-Conflict labor-allocation game: for every $i \in \mathcal{N}$,

$$\pi_i^C(L_i^*, L^*_{-i}) \geq \pi_i^C(L_i, L^*_{-i}) \qquad \forall\, L_i \in \mathcal{L}_i.$$

Every such strategy profile is a subgame perfect equilibrium of the theft and trade game.

Proof. I show that this strategy profile exists and is an SPE in three steps. Step 1 shows that the continuation rule ρ^* reduces the full game to the induced Conflict game over labor allocations. Step 2 establishes the existence of an equilibrium of the induced Conflict game using Reny (1999). Step 3 verifies that the resulting strategy profile is an SPE of the full theft and trade game.

Step 1: Inducing a game over labor allocations. Under ρ^*, every player declares theft toward every other player at every $L \in \mathcal{L}$. Every pair of players is therefore joined by a Conflict edge, and the interaction graph has a single component: a universal Conflict comprising all of $\mathcal{N}$. Because no candidate contains players joined by a Conflict

edge, every admissible coalition with two or more members dissolves and every candidate is a singleton, and each player n receives her universal-Conflict payoff $\pi_n^C(L)$. The labor-allocation stage therefore reduces to the universal-Conflict labor-allocation game, with strategy space $\mathcal{L}$ and payoff functions $\{\pi_n^C\}_{n\in\mathcal{N}}$. The next step establishes that this induced game has an equilibrium.

Step 2: Existence of a NE on the induced game. To apply Reny's existence theorem to the induced Conflict game, the action space must be non-empty, convex and compact, the payoff function must be quasiconcave and bounded, and the game must be "better-reply secure"[13].

Player n's feasible labor-allocation set $\mathcal{L}_n$ is a simplex, and therefore nonempty, compact, and convex. Hence, the action space for the induced game, $\mathcal{L} = \prod_{i\in\mathcal{N}} \mathcal{L}_i$ is nonempty, compact, and convex.

Since $U\big(Y(\mathcal{N})\big)$ is continuous in L and $\mathcal{L}$ is compact, $U\big(Y(\mathcal{N})\big)$ is bounded on $\mathcal{L}$. Since $0 \le q_n \le 1$, we have $0 \le \pi_n^C \le U\big(Y(\mathcal{N})\big)$, so payoffs in the induced Conflict game are bounded. Bounded payoffs combined with the nonempty, compact, convex action sets $\mathcal{L}_n$ make the induced Conflict game compact in Reny's sense. Next I show that each player's payoff in the induced game is quasiconcave in her own labor allocation.

Fix the labor allocations of all players other than n in the induced Conflict game, and let $S_{-n} = \sum_{j\neq n} S_j$ be their total security output. Since U is increasing with $U(\mathbf{0}) = 0$, $U\big(Y(\mathcal{N})\big)$ is nonnegative, and by Lemma C.3 (Concavity of Utility in Labor) it is concave in t_n.

The other factor of the payoff, $q_n(s_n)$, is nonnegative everywhere and concave in s_n for every $s_n > 0$. If $S_{-n} > 0$, Lemma C.1 (Concavity of the Contest Share) shows that q_n, which equals p_n in the universal Conflict, is concave. If $S_{-n} = 0$, then $q_n(s_n) = 1$ for every $s_n > 0$, so q_n is constant and therefore weakly concave.

The discontinuity at $s_n = 0$ requires separate treatment. First consider the region

$$P = \left\{L_n \in \mathcal{L}_n : s_n > 0,\ U\big(Y(\mathcal{N})\big) > 0\right\}.$$

The set P is convex. The condition $s_n > 0$ is preserved under convex combinations, as is positivity of the concave function $U\big(Y(\mathcal{N})\big)$. On P, both q_n and $U\big(Y(\mathcal{N})\big)$ are positive and concave. Their logarithms are therefore concave, so

$$\ln \pi_n^C = \ln q_n + \ln U\big(Y(\mathcal{N})\big)$$

is concave on P. Hence π_n^C is quasiconcave on P.

Quasiconcavity on P extends immediately to any segment in $\mathcal{L}_n$ with a zero-payoff endpoint. Consider any two allocations $L_n^{'}, L_n^{''} \in \mathcal{L}_n$. If either endpoint has zero payoff, then for every $\lambda \in [0,1]$,

$$\pi_n^C(\lambda L_n^{'} + (1-\lambda)L_n^{''}) \ge 0 = \min\{\pi_n^C(L_n^{'}), \pi_n^C(L_n^{''})\},$$

so the quasiconcavity condition holds immediately.

The only cases remaining have positive payoffs at both endpoints and $s_n = 0$ at at least one endpoint. A positive payoff at $s_n = 0$ requires $S_{-n} = 0$, since otherwise $q_n(0) = 0$.

[13]In this context, secure is used in Reny's original sense and is unrelated to security goods in the model

At every interior point of the segment, $s_n > 0$, so $q_n = 1$ and

$$\pi_n^C = U\big(Y(\mathcal{N})\big).$$

Concavity of $U\big(Y(\mathcal{N})\big)$ therefore gives

$$U\left(X_n^P(\lambda t_n^{'} + (1-\lambda)t_n^{''}) + \sum_{i\neq n} X_i^P(t_i)\right) \geq \min\left\{U\left(X_n^P(t_n^{'}) + \sum_{i\neq n} X_i^P(t_i)\right), U\left(X_n^P(t_n^{''}) + \sum_{i\neq n} X_i^P(t_i)\right)\right\}.$$

Since $q_n \leq 1$ at both endpoints, the right-hand side is at least

$$\min\{\pi_n^C(L_n^{'}), \pi_n^C(L_n^{''})\}.$$

The last case has $q_n = \frac{1}{N}$ everywhere on the line segment between endpoints, so π_n^C is concave, and therefore quasiconcave. Thus π_n^C is quasiconcave in L_n on $\mathcal{L}_n$.

Since the payoff function is continuous whenever $s \neq \mathbf{0}$, better-reply security follows directly at every such profile. Therefore better-reply security only needs to be verified at $s = \mathbf{0}$. Fix an arbitrary sequence $L^k \to L^0$ with $s^0 = \mathbf{0}$. Since $\pi^C(L^k)$ is bounded, it has a convergent subsequence by Bolzano-Weierstrass. Because different subsequences approaching L^0 may generate different limiting payoff vectors, take w to be the limit of an arbitrary convergent subsequence and relabel this subsequence as L^k.

Because Conflict divides a common aggregate bundle proportionally and utility is homogeneous of degree one, the limiting individual payoffs sum to the utility of aggregate production:

$$\sum_{i\in\mathcal{N}} w_i = U\left(\sum_{i\in\mathcal{N}} X_i^P(t_i^0)\right).$$

If

$$U\left(\sum_{i\in\mathcal{N}} X_i^P(t_i^0)\right) > 0,$$

there exists some player n such that

$$w_n < U\left(\sum_{i\in\mathcal{N}} X_i^P(t_i^0)\right).$$

Fix such a player n. If instead aggregate utility is zero, then $w_i = 0$ for every $i \in \mathcal{N}$; fix an arbitrary player n. If $w_n = 0$, choose any $0 < \hat{s}_n < \tau$ and set

$$\hat{t}_{nm} = \frac{\tau - \hat{s}_n}{M} \qquad \forall m \in \mathcal{M}.$$

Since every good has positive aggregate production, utility is strictly positive.

If $w_n > 0$, choose $\hat{s}_n > 0$ sufficiently small that the proportional reduction

$$\hat{t}_{nm} = \left(1 - \frac{\hat{s}_n}{\tau}\right) t_{nm}^0 \qquad \forall m \in \mathcal{M}$$

preserves

$$U\left(X_n^P(\hat{t}_n) + \sum_{i \neq n} X_i^P(t_i^0)\right) > w_n.$$

Such an $\hat{L}_n = (\hat{t}_n, \hat{s}_n)$ exists by continuity of production and utility, since n was chosen so that aggregate utility at L^0 strictly exceeds w_n.

In both cases, since $s_{-n}^0 = \mathbf{0}$ and $\hat{s}_n > 0$, $q_n(\hat{s}_n, s_{-n}^0) = 1$. Therefore,

$$\pi_n^C(\hat{L}_n, L_{-n}^0) = U\left(X_n^P(\hat{t}_n) + \sum_{i \neq n} X_i^P(t_i^0)\right) > w_n.$$

Because π_n^C is continuous whenever $s \neq \mathbf{0}$, there exist $\delta > 0$ and a neighborhood V_{-n} of L_{-n}^0 such that

$$\pi_n^C(\hat{L}_n, L_{-n}) > w_n + \delta \qquad \forall\, L_{-n} \in V_{-n}.$$

Player n can therefore secure a payoff strictly greater than w_n. Since the convergent subsequence and its limiting payoff vector w were arbitrary, the Conflict game is better-reply secure at $s = \mathbf{0}$, and therefore better-reply secure everywhere.

The Conflict game consequently satisfies the conditions of Theorem 3.1 of Reny (1999): strategy sets are nonempty, compact, and convex; payoffs are bounded and quasiconcave in each player's own strategy; and the game is better-reply secure. It therefore possesses a pure-strategy Nash equilibrium

$$L^* = (L_i^*)_{i \in \mathcal{N}},$$

which satisfies condition (ii).

Step 3: No profitable deviation in any subgame. Fix a player $n \in \mathcal{N}$ and any unilateral deviation from $\sigma_n^* = (L_n^*, \rho^*)$—to a labor allocation $\hat{L}_n \in \mathcal{L}_n$, a posture declaration, an alternative contingent action rule, or any combination. Whatever she chooses, the realized labor profile is $(\hat{L}_n, L_{-n}^*)$, and because every other player is still invoking ρ^*, each player declares theft toward n at this profile, regardless of what n declares. Thus, n remains a combatant in the Conflict component exhausting $\mathcal{N}$, with payoff $\pi_n^C(\hat{L}_n, L_{-n}^*)$, which by condition (ii) is at most $\pi_n^C(L_n^*, L_{-n}^*)$. Therefore, the strategy profile σ^* is a Nash equilibrium of the full theft and trade game.

Any individual player's posture choices are payoff-irrelevant given the others' theft declarations, so $\rho^*(L)$ is a Nash equilibrium of the posture subgame following L, for every $L \in \mathcal{L}$. The profile $\sigma^* = (L^*, \rho^*)$ induces a Nash equilibrium in every subgame, and hence a subgame perfect equilibrium.

■

Corollary 1 ***Law of One Power Price*****:** *Let σ^* be an SPE, and for any component H of $G(\sigma^*)$*

let $q_H \equiv \sum_{n \in V(H)} q_n(s^*)$ *denote its members' combined power. Then*

$$Y(H) = q_H\, Y(\mathcal{N}) \qquad \textit{and} \qquad U\big(Y(H)\big) = q_H\, U\big(Y(\mathcal{N})\big).$$

Every component's consumer-good production is the same fraction of aggregate production as its members' combined power, and its members realize exactly that share of aggregate utility.

Proof. By Theorem 1 (Characterization), L^* is a Nash equilibrium of the universal-Conflict labor-allocation game, so every player holds positive security by Proposition 4 (Security Budget); hence $q_n(s^*) > 0$ for every n and $q_H > 0$ for every component H. Combining Theorem 1 with Lemma 1 (Aggregate Payoff Identity),

$$U\big(Y(H)\big) = \sum_{n \in V(H)} \pi_n(\sigma^*) = q_H\, U\big(Y(\mathcal{N})\big) > 0$$

for every component H, which is the second identity. Since every consumer good is essential, each $Y(H)$ lies in $\mathbb{R}^M_{++}$. Summing over the components, which partition $\mathcal{N}$, gives $\sum_H U\big(Y(H)\big) = U\big(Y(\mathcal{N})\big)$, so the inequality of Lemma C.5 (Superadditivity of U) holds with equality.

Equality requires the bundles $Y(H)$ to be proportional. For any $x, y \in \mathbb{R}^M_{++}$, write $\hat{x} = x/U(x)$, $\hat{y} = y/U(y)$ and $\lambda = U(x)/\big(U(x) + U(y)\big)$. By homogeneity, $U(\hat{x}) = U(\hat{y}) = 1$ and

$$U(x + y) = \big(U(x) + U(y)\big)\, U\big(\lambda \hat{x} + (1 - \lambda)\hat{y}\big).$$

If $\hat{x} \neq \hat{y}$, strict quasiconcavity gives $U\big(\lambda \hat{x} + (1 - \lambda)\hat{y}\big) > 1$, and $U(x + y) > U(x) + U(y)$. Applying this pairwise, equality in Lemma C.5 requires $\hat{Y}(H)$ to be the same for every component, so $Y(H) = c_H\, Y(\mathcal{N})$ for scalars $c_H > 0$. Homogeneity then gives $U\big(Y(H)\big) = c_H\, U\big(Y(\mathcal{N})\big)$, and the second identity gives $c_H = q_H$.

■

Corollary 2 ***Law of One Price*****:** *In any SPE, all Markets clear at the same relative prices. Moreover, the common price vector is proportional to* $\nabla U\big(Y(\mathcal{N})\big)$.

Proof. Fix a Market D. Because preferences are common and homothetic, demand aggregates to that of a single consumer with utility U, and the unique market-clearing price vector supports the Market's aggregate bundle: $p \propto \nabla U\big(Y(D)\big)$, which is well defined because $Y(D) \in \mathbb{R}^M_{++}$ by Corollary 1 (Law of One Power Price). By the same corollary, $Y(D) = q_D\, Y(\mathcal{N})$ with $q_D > 0$. Since U is homogeneous of degree one, ∇U is homogeneous of degree zero, so $\nabla U\big(Y(D)\big) = \nabla U\big(Y(\mathcal{N})\big)$.

■

Corollary 3 ***Autarky Exclusion*****:** *If an SPE contains an Autarkic player n, then for every pair of consumption goods m, m',*

$$\frac{f'_{nm}(t_{nm})}{f'_{nm'}(t_{nm'})} = \frac{\partial U(Y(\mathcal{N}))/\partial X_{m'}}{\partial U(Y(\mathcal{N}))/\partial X_m}.$$

Thus, Autarky can occur in equilibrium only if no gains from unilateral specialization remain for that player.

Proof. Given an arbitrary theft and trade game, if SPE σ^* has some player n who is autarkic, by Theorem 1 (Characterization), t_n^* maximizes $U(Y(\mathcal{N}))$ with respect to t_n, holding security allocations at s^* and all other players' labor allocations fixed. Thus, for every pair of consumption goods m, m', the interior first-order conditions imply

$$\frac{f'_{nm}(t^*_{nm})}{f'_{nm'}(t^*_{nm'})} = \frac{\partial U(Y(\mathcal{N}))/\partial X_{m'}}{\partial U(Y(\mathcal{N}))/\partial X_m}.$$

By Corollary 1 (Law of One Power Price), $X_n^P(t_n^*)$ is proportional to $Y(\mathcal{N})$. Since preferences are homothetic, the marginal rates of substitution are identical at both bundles. Consequently, t_n^* also satisfies the first-order conditions for maximizing $U(X_n^P(t_n))$.

■

Proposition 3 ***Common Labor Allocation*****:** *Fix a security allocation s with $0 < s_i \leq \tau$ for every $i \in \mathcal{N}$ and $s_i < \tau$ for at least one i. Consider the production games induced by a single connected Market and a single universal Conflict among all players in $\mathcal{N}$. The two games have the same unique Nash equilibrium in consumption-good labor allocations:*

$$t^{T*} = t^{C*}.$$

Proof. Fix s as specified, and consider the joint problem

$$\max_t \; U\Big(\sum_{n\in\mathcal{N}} X_n^P(t_n)\Big) \quad \text{subject to} \quad \sum_{m=1}^{M} t_{nm} = \tau - s_n, \;\; t_{nm} \geq 0 \quad \forall\, n \in \mathcal{N},\; m \in \mathcal{M}. \tag{16}$$

A player with $s_n = \tau$ has the single feasible allocation $t_n = \mathbf{0}$.

Conflict. Security is fixed, so each player chooses how to divide $\tau - s_n$ among consumer goods. Every player's payoff is $q_n(s)\,U\big(Y(\mathcal{N})\big)$, and since $s_n > 0$ for every n, $q_n(s)$ is a positive constant. The induced game is therefore an identical-interest game with common payoff $U\big(Y(\mathcal{N})\big)$, and any solution of (16) is a Nash equilibrium.

The feasible set of (16) is nonempty and compact and the objective is continuous, so by the Weierstrass theorem it attains a maximum U^*. Since some player has $s_n < \tau$, she can allocate positive labor to every good, so $U^* > 0$.

Conversely, let t be a Nash equilibrium. Then $U\big(Y(\mathcal{N})\big) > 0$; otherwise a player with $s_n < \tau$ could allocate positive labor to every good and make it positive. Every component of $Y(\mathcal{N})$ is therefore positive because every consumer good is essential, and $\partial U\big(Y(\mathcal{N})\big)/\partial X_m > 0$ for every m by Lemma C.4 (Positive Marginal Utility). At t, no player with $s_n < \tau$ allocates zero labor to a good whose production function has an infinite right derivative at zero: shifting a small amount of labor into it from a good she produces would raise $U\big(Y(\mathcal{N})\big)$ at an infinite rate against a finite loss. Let g_n denote the gradient of $U\big(Y(\mathcal{N})\big)$ with respect to t_n at t, using right derivatives where labor is zero; each g_n is therefore finite. Because $U\big(Y(\mathcal{N})\big)$ depends on labor only through $Y(\mathcal{N})$, its one-sided directional derivative at t in any feasible direction $d = (d_n)_{n\in\mathcal{N}}$ is $\sum_n g_n \cdot d_n$, and since each t_n is a best response, $g_n \cdot d_n \le 0$. By Lemma C.3 (Concavity of Utility in Labor), $U\big(Y(\mathcal{N})\big)$ is concave in t, so for any feasible t',

$$U\big(Y(\mathcal{N})\big)(t') \le U\big(Y(\mathcal{N})\big)(t) + \sum_{n\in\mathcal{N}} g_n \cdot (t'_n - t_n) \le U\big(Y(\mathcal{N})\big)(t).$$

Hence t solves (16), and a profile is a Nash equilibrium if and only if it solves (16).

Trade. Every Trade equilibrium solves (16). Let t be the production profile of a Trade equilibrium with prices p. Each player's t_n maximizes the market value of her production, and t^*_n is feasible for her, so

$$\sum_{m=1}^{M} p_m f_{nm}(t^*_{nm}) \;\le\; \sum_{m=1}^{M} p_m f_{nm}(t_{nm}) \qquad \forall\, n \in \mathcal{N}.$$

Summing over players, the bundle $\sum_{n\in\mathcal{N}} X^P_n(t^*_n)$ costs no more than $\sum_{n\in\mathcal{N}} X^P_n(t_n)$ at prices p. Preferences are common and U is homogeneous of degree one, so scaling a budget scales the optimal bundle. Each player's consumption bundle X^F_n is therefore a scaled copy of a single bundle, and by market clearing their sum $\sum_{n\in\mathcal{N}} X^P_n(t_n)$ maximizes U over all bundles costing no more than $\sum_{n\in\mathcal{N}} X^P_n(t_n)$. Since $\sum_{n\in\mathcal{N}} X^P_n(t^*_n)$ is such a bundle,

$$U\Big(\sum_{n\in\mathcal{N}} X^P_n(t^*_n)\Big) \;\le\; U\Big(\sum_{n\in\mathcal{N}} X^P_n(t_n)\Big),$$

so t solves (16).

A Trade equilibrium exists: the joint maximizer t^*, with prices equal to the marginal utilities at its aggregate production, satisfies every equilibrium condition.[14] At t^*, ag-

[14]This is the Second Welfare Theorem specialized to common homothetic preferences: because aggregate demand does not depend on the distribution of wealth (Gorman 1953), the planner's production plan is supported without transfers.

gregate production is strictly positive because $U^* > 0$ and every consumer good is essential. Set each price p_m equal to the marginal utility $\partial U(Y(\mathcal{N}))/\partial f_{nm}(t^*_{nm})$, which is the same for every n because $U(Y(\mathcal{N}))$ depends on $f_{nm}(t_{nm})$ only through aggregate production of good m.

- *Consumers.* At these prices, $Y(\mathcal{N})$ satisfies the consumer optimality conditions, so by concavity of U it maximizes U over all bundles costing no more than aggregate production. Every price is positive: if some good had a zero price, adding a unit of it to $Y(\mathcal{N})$ would leave the bundle affordable without lowering utility, contradicting uniqueness of the optimal bundle under strict quasiconcavity. By homogeneity and strict quasiconcavity, each player's unique demand is the multiple of $Y(\mathcal{N})$ whose value equals the market value of her own production. A player with $s_n = \tau$ produces nothing, and since every price is positive, the only bundle she can afford is the zero bundle.
- *Producers.* The KKT conditions are necessary at t^* because the constraints of (16) are linear. For every n and every pair of goods m, m' with $t^*_{nm} > 0$,

$$\frac{\partial}{\partial t_{nm}} U\Big(\sum_{i\in\mathcal{N}} X^P_i(t^*_i)\Big) \;\geq\; \frac{\partial}{\partial t_{nm'}} U\Big(\sum_{i\in\mathcal{N}} X^P_i(t^*_i)\Big).$$

  By the chain rule, this is

$$p_m \,\frac{df_{nm}(t^*_{nm})}{dt_{nm}} \;\geq\; p_{m'}\,\frac{df_{nm'}(t^*_{nm'})}{dt_{nm'}}.$$

  Since prices are positive, the market value of player n's production is concave over her simplex, and these conditions are sufficient for t^*_n to maximize it.
- *Clearing.* The market values of the players' production sum to the value of aggregate production, so aggregate demand equals $Y(\mathcal{N})$ and the Market clears.

Uniqueness. Suppose two distinct feasible allocations $t' \neq t''$ both attain U^*, and let $\bar{t} = \frac{1}{2}t' + \frac{1}{2}t''$, which is feasible by convexity. Concavity of each f_{nm} gives, componentwise,

$$\sum_{n\in\mathcal{N}} X^P_n(\bar{t}_n) \;\geq\; \tfrac{1}{2}\sum_{n\in\mathcal{N}} X^P_n(t'_n) + \tfrac{1}{2}\sum_{n\in\mathcal{N}} X^P_n(t''_n). \tag{17}$$

- If $\sum_{n\in\mathcal{N}} X^P_n(t'_n) \neq \sum_{n\in\mathcal{N}} X^P_n(t''_n)$, then (17), monotonicity of U, and strict quasiconcavity of U give

$$U\Big(\sum_{n\in\mathcal{N}} X^P_n(\bar{t}_n)\Big) \;\geq\; U\Big(\tfrac{1}{2}\sum_{n\in\mathcal{N}} X^P_n(t'_n) + \tfrac{1}{2}\sum_{n\in\mathcal{N}} X^P_n(t''_n)\Big) \;>\; U^*,$$

  which contradicts optimality.
- If $\sum_{n\in\mathcal{N}} X^P_n(t'_n) = \sum_{n\in\mathcal{N}} X^P_n(t''_n)$, then since $t' \neq t''$ there exist n and m with $t'_{nm} \neq t''_{nm}$. Strict concavity of f_{nm} makes (17) strict in good m, so

$$\sum_{i\in\mathcal{N}} X^P_i(\bar{t}_i) \;\geq\; \sum_{i\in\mathcal{N}} X^P_i(t'_i)$$

  componentwise, with strict inequality in good m. By monotonicity of U, $\bar{t}$ also attains U^*, yet its aggregate bundle differs from that of t'. The first case then

applies to the pair $t', \bar{t}$ and gives a contradiction.

Hence t^* is the unique solution to (16), and by the Conflict step it is the unique Nash equilibrium of the universal Conflict game.

Because both induced games therefore have the same unique equilibrium:

$$t^{T*} = t^{C*} = t^*.$$

■

Proposition 4 ***Security Budget*****:** *Let L^* be a Nash equilibrium of the universal-Conflict labor-allocation game. Every player holds positive security, $s_n^* > 0$, and for every consumer good m,*

$$1 - q_n(s^*) \;\geq\; \frac{\partial \ln U\big(Y(\mathcal{N})\big)/\partial t_{nm}}{d \ln f_{nS}(s_n^*)/ds_n}, \qquad \textit{with equality if } t_{nm}^* > 0,$$

where derivatives at $t_{nm} = 0$ are right derivatives.

Proof. Fix a player n and hold every other player's labor at L_{-n}^*. Player n's payoff in the universal Conflict is $q_n\, U\big(Y(\mathcal{N})\big)$, with $q_n = S_n/(S_n + S_{-n})$ and $S_{-n} = \sum_{j \neq n} S_j$.

Positive aggregate utility. $U\big(Y(\mathcal{N})\big) > 0$ at L^*. Otherwise every payoff is zero, and any player could hold a small amount of security, allocate her remaining labor to every consumer good, and obtain a positive payoff. Hence every component of $Y(\mathcal{N})$ is positive because every consumer good is essential, U is differentiable at $Y(\mathcal{N})$, and $\partial U\big(Y(\mathcal{N})\big)/\partial X_m > 0$ for every m by Lemma C.4 (Positive Marginal Utility).

Positive security. Suppose $s_n^* = 0$. If $S_{-n} > 0$, then $q_n = 0$ and n's payoff is zero; shifting a small amount of labor from a good she produces into security makes q_n positive and, by continuity, keeps $U\big(Y(\mathcal{N})\big)$ positive, a profitable deviation. If $S_{-n} = 0$, no player holds security, $q_n = 1/N$ by (13), and the same shift raises q_n to 1 while, by continuity, keeping $U\big(Y(\mathcal{N})\big)$ arbitrarily close to its original value, a profitable deviation since $N \geq 2$. Hence $s_n^* > 0$ for every player, and therefore $S_{-n} > 0$.

Marginal return to security. Since $s_n^* > 0$ and f_{nS} is C^2 at positive labor, q_n is differentiable at s_n^*, and

$$\frac{d \ln q_n}{ds_n} = \frac{d \ln f_{nS}}{ds_n} - \frac{d \ln\big(f_{nS} + S_{-n}\big)}{ds_n} = (1 - q_n)\frac{d \ln f_{nS}(s_n^*)}{ds_n},$$

which is positive and finite because f_{nS} is increasing and strictly concave.

Finite marginal products. Suppose $t_{nm}^* = 0$ for a good m with $f'_{nm}(0^+) = \infty$. Moving ε units of labor from security to good m raises the output of good m by $f_{nm}(\varepsilon)$, with $f_{nm}(\varepsilon)/\varepsilon \to \infty$, and lowers $\ln q_n$ by at most a finite multiple of ε for small ε. Since $\ln U\big(Y(\mathcal{N})\big)$ is differentiable at $Y(\mathcal{N})$ with $\partial \ln U/\partial X_m > 0$, the change in n's log payoff

divided by ε tends to $+\infty$, a profitable deviation for small ε. Hence every good whose production function has an infinite right derivative at zero is produced by every player, and at L^* the right derivative $\partial \ln U(Y(\mathcal{N}))/\partial t_{nm}$ is finite for every m.

Optimality conditions. Because L_n^* is a best response and n's payoff is positive, her log payoff $\ln q_n + \ln U(Y(\mathcal{N}))$ cannot rise along any feasible direction. Since $s_n^* > 0$, moving labor from security to good m is feasible for every m, and its right derivative must be nonpositive:

$$\frac{\partial \ln U(Y(\mathcal{N}))}{\partial t_{nm}} - \frac{d \ln q_n}{ds_n} \leq 0.$$

If $t_{nm}^* > 0$, moving labor from good m to security is also feasible, giving the reverse inequality, so equality holds. Substituting the marginal return to security and dividing by $d \ln f_{nS}(s_n^*)/ds_n > 0$ gives

$$1 - q_n(s^*) \;\geq\; \frac{\partial \ln U(Y(\mathcal{N}))/\partial t_{nm}}{d \ln f_{nS}(s_n^*)/ds_n}, \qquad \text{with equality if } t_{nm}^* > 0.$$

Maximum form. If $s_n^* < \tau$, then $t_{nm}^* > 0$ for some good m, which attains the bound with equality, while every other good satisfies the inequality. Hence $1 - q_n(s^*)$ equals the maximum of the ratio over $\mathcal{M}$.

■

Lemma C.6 ***Value Bound*:** *Fix a labor profile at which every component of $Y(\mathcal{N})$ is positive. For any set of players $I \subseteq \mathcal{N}$,*

$$U(Y(I)) \;\leq\; \sum_{i \in I} \sum_{m=1}^{M} \frac{\partial U(Y(\mathcal{N}))}{\partial f_{im}(t_{im})} f_{im}(t_{im}),$$

with equality when $Y(I)$ is a scalar multiple of $Y(\mathcal{N})$.

Proof. Since U is concave (Lemma C.2 (Concavity of U)) and differentiable at $Y(\mathcal{N})$, it lies below its tangent plane there. Because $U(Y(\mathcal{N}))$ depends on $f_{im}(t_{im})$ only through aggregate production of good m, the marginal utility $\partial U(Y(\mathcal{N}))/\partial f_{im}(t_{im})$ is the same for every i, and the tangent-plane inequality reads

$$U(Y(I)) \;\leq\; U(Y(\mathcal{N})) + \sum_{m=1}^{M} \frac{\partial U(Y(\mathcal{N}))}{\partial f_{nm}(t_{nm})} \Big(\sum_{i \in I} f_{im}(t_{im}) - \sum_{i \in \mathcal{N}} f_{im}(t_{im}) \Big).$$

Because U is homogeneous of degree one, Euler's theorem gives

$$\sum_{i \in \mathcal{N}} \sum_{m=1}^{M} \frac{\partial U(Y(\mathcal{N}))}{\partial f_{im}(t_{im})} f_{im}(t_{im}) = U(Y(\mathcal{N})),$$

so the terms involving $Y(\mathcal{N})$ cancel, leaving the stated inequality. If $Y(I) = cY(\mathcal{N})$, homogeneity gives $U\big(Y(I)\big) = cU\big(Y(\mathcal{N})\big)$, and Euler's theorem shows the right-hand side also equals $cU\big(Y(\mathcal{N})\big)$.

■

Proposition 5 ***Conditions for Social Order*****:** *A Social Order equilibrium exists if and only if there exist a Nash equilibrium L^* of the universal-Conflict labor-allocation game and a set $I \subseteq \mathcal{N}$ with $|I| \geq 2$ such that, at L^*,*

(i) $Y(I) = \Big(\sum_{i\in I} q_i(s^*)\Big)\, Y(\mathcal{N})$, *and*

(ii) when the players of I form a Market, $\alpha_i = \dfrac{q_i(s^*)}{\sum_{j\in I} q_j(s^*)}$ *for every $i \in I$.*

Proof. Throughout, candidates trading in a Market are singletons. Fix a labor profile $L^* = (t^*, s^*)$ that is a Nash equilibrium of the universal-Conflict labor-allocation game, and write $S_n = f_{nS}(s^*_n)$ and $q_n = q_n(s^*) = S_n / \sum_{j\in\mathcal{N}} S_j$.

Step 1: Positive production at L^*. By Proposition 4 (Security Budget), $q_n > 0$ for every n. If $U\big(Y(\mathcal{N})\big) = 0$, every payoff in the universal Conflict is zero, and any player could hold a small amount of security, allocate her remaining labor to every consumer good, and obtain a positive payoff. Hence $U\big(Y(\mathcal{N})\big) > 0$, and since every consumer good is essential, every component of $Y(\mathcal{N})$ is positive.

Step 2: Prices and values. For each good m, let p_m be the marginal utility $\partial U\big(Y(\mathcal{N})\big)/\partial f_{nm}(t^*_{nm})$, which is the same for every n. By Euler's theorem,

$$\sum_{n\in\mathcal{N}} \sum_{m=1}^{M} p_m f_{nm}(t^*_{nm}) = U\big(Y(\mathcal{N})\big). \tag{18}$$

Let $I \subseteq \mathcal{N}$ with $|I| \geq 2$ and suppose $Y(I)$ is a scalar multiple of $Y(\mathcal{N})$. If the players of I form a Market, every member's demand is proportional to $Y(I)$, so each member's bundle is interior, and market-clearing price ratios equal marginal rates of substitution at $Y(I)$. Because U is homogeneous of degree one, its partial derivatives are homogeneous of degree zero, so these ratios equal those of p. Each member's budget is the value of her production, so her share of $Y(I)$ is her share of its value:

$$\alpha_n = \frac{\sum_m p_m f_{nm}(t^*_{nm})}{\sum_{i\in I}\sum_m p_m f_{im}(t^*_{im})} \qquad \forall\, n \in I. \tag{19}$$

Step 3: Conditions (i) and (ii) are equivalent to alignment of every member of I. Suppose (i) holds. By homogeneity, $U\big(Y(I)\big) = \big(\sum_{i\in I} q_i\big) U\big(Y(\mathcal{N})\big)$, and by the equality case of Lemma C.6 (Value Bound),

$$\sum_{i\in I}\sum_m p_m f_{im}(t^*_{im}) = U\big(Y(I)\big) = \Big(\sum_{i\in I} q_i\Big) U\big(Y(\mathcal{N})\big).$$

Substituting into (19), condition (ii) holds if and only if

$$\sum_{m=1}^{M} p_m f_{nm}(t^*_{nm}) = q_n\, U\big(Y(\mathcal{N})\big) \qquad \forall\, n \in I. \tag{20}$$

Moreover, the complement $R = \mathcal{N} \setminus I$ satisfies $Y(R) = Y(\mathcal{N}) - Y(I) = \big(\sum_{j\in R} q_j\big) Y(\mathcal{N})$, since the q_j sum to one. By (18),

$$\sum_{j\in R} \sum_{m} p_m f_{jm}(t^*_{jm}) = \Big(\sum_{j\in R} q_j\Big) U\big(Y(\mathcal{N})\big) = U\big(Y(R)\big). \tag{21}$$

Necessity. Let σ^* be a Social Order equilibrium. Since $E^T(\sigma^*) \neq \emptyset$, $G(\sigma^*)$ has a Market component D; let $I = V(D)$, so $|I| \geq 2$. By Theorem 1 (Characterization), the equilibrium labor profile L^* is a Nash equilibrium of the universal-Conflict labor-allocation game and $\pi_n(\sigma^*) = q_n\, U\big(Y(\mathcal{N})\big)$ for every n. By Corollary 1 (Law of One Power Price), $Y(I) = c\, Y(\mathcal{N})$ for some $c \geq 0$. By Lemma 1 (Aggregate Payoff Identity) and homogeneity,

$$c\, U\big(Y(\mathcal{N})\big) = U\big(Y(I)\big) = \sum_{n\in I} \pi_n(\sigma^*) = \Big(\sum_{n\in I} q_n\Big) U\big(Y(\mathcal{N})\big),$$

so $c = \sum_{n\in I} q_n$ by Step 1, which is condition (i). Each member's payoff is $\alpha_n U\big(Y(I)\big) = q_n U\big(Y(\mathcal{N})\big)$, so $\alpha_n = q_n / \sum_{i\in I} q_i$, which is condition (ii).

Sufficiency. Let L^* and I satisfy (i) and (ii), and let $R = \mathcal{N} \setminus I$. By Step 3, every member of I satisfies (20). Without loss of generality (see the discussion preceding Proposition 5 (Conditions for Social Order)), if $|R| = 1$ its single member is autarkic. Define $\hat{\rho}$ as follows, with every candidate a singleton.

- At $L = L^*$: each $n \in I$ declares trade toward every other member of I and none toward R. If $|R| \geq 2$, each $j \in R$ declares theft toward every other member of R and none toward I; if $|R| = 1$, its member declares none toward every player.
- At any $L \neq L^*$: every player declares theft toward every other player.

Let $\hat{\sigma} = (L^*, \hat{\rho})$. On the path, I is a Market with a complete trade graph and R is a Conflict, or an autarkic singleton. Since $E^T(\hat{\sigma}) \neq \emptyset$, it remains to show that $\hat{\sigma}$ is an SPE.

Payoffs on the path. By (i), (ii), and homogeneity, each $n \in I$ receives $\alpha_n U\big(Y(I)\big) = \frac{q_n}{\sum_{i\in I} q_i}\big(\sum_{i\in I} q_i\big) U\big(Y(\mathcal{N})\big) = q_n U\big(Y(\mathcal{N})\big)$. Each $j \in R$ wins the Conflict over $Y(R)$ with probability $S_j / \sum_{k\in R} S_k = q_j / \sum_{k\in R} q_k$, and $U\big(Y(R)\big) = \big(\sum_{k\in R} q_k\big) U\big(Y(\mathcal{N})\big)$, so she also receives $q_j U\big(Y(\mathcal{N})\big)$. If R is a singleton, its member receives $U\big(Y(R)\big) = q_j U\big(Y(\mathcal{N})\big)$ directly. Every player's on-path payoff is therefore $q_n U\big(Y(\mathcal{N})\big)$, her payoff in the universal Conflict at L^*.

Subgames after $L \neq L^*$. Every player declares theft toward every other player, so each player is a combatant whatever postures she chooses. As in Step 3 of the proof of Proposition 2 (Existence of Warre SPE), no posture deviation changes any payoff, and universal theft is a Nash equilibrium of every such subgame.

Round-one deviations. If player n deviates to $L_n \neq L_n^*$, universal theft follows, and her payoff is her universal-Conflict payoff at (L_n, L_{-n}^*). Since L^* is a Nash equilibrium of the universal-Conflict labor-allocation game, this is at most her universal-Conflict payoff at L^*, which equals her on-path payoff.

Posture deviations at L^*. Fix player n and any change in her postures, holding all other postures fixed.

- Suppose n declares no theft. If $n \in I$, the other members of I still trade with one another along a complete graph. If n keeps at least one trade posture toward I, the Market is unchanged. If she keeps none, she is autarkic and receives $U(X_n^P)$, which by Lemma C.6 with $I = \{n\}$ and (20) is at most $q_n U(Y(\mathcal{N}))$. If $n \in R$ and $|R| \geq 2$, the other members of R still declare theft toward her, so the Conflict over $Y(R)$ is unchanged. No player declares trade toward a player in the other set, so no new trade edge can form.
- Suppose n declares theft toward a nonempty set of players. Then n is a combatant. Let J be the vertex set of her Conflict component after the deviation, and $S_J = \sum_{k \in J} S_k$. Her payoff is $(S_n/S_J)\,U(Y(J))$. Every player she targets in I becomes a combatant in J. If $n \in R$, or if n targets any member of R, then all of R lies in J, since the members of R remain connected by their theft declarations. So J is the union of a subset of I and either all of R or none of it. By Lemma C.6, (20), and (21),

$$U(Y(J)) \;\leq\; \sum_{k \in J} \sum_{m} p_m f_{km}(t_{km}^*) = \Big(\sum_{k \in J} q_k\Big) U(Y(\mathcal{N})) = \frac{S_J}{\sum_{j \in \mathcal{N}} S_j}\, U(Y(\mathcal{N})),$$

  so her payoff is at most $\frac{S_n}{S_J} \cdot \frac{S_J}{\sum_j S_j} U(Y(\mathcal{N})) = q_n U(Y(\mathcal{N}))$.

No posture deviation is profitable, so $\hat{\rho}(L^*)$ is a Nash equilibrium of the subgame at L^*. Hence $\hat{\sigma}$ is a subgame-perfect equilibrium with $E^T(\hat{\sigma}) \neq \emptyset$, a Social Order equilibrium.

■

Corollary 4 *Conditions for Trade*: *A Trade equilibrium exists if and only if, at some Nash equilibrium L^* of the universal-Conflict labor-allocation game, $\alpha_i = q_i(s^*)$ for every $i \in \mathcal{N}$ when all players form a single Market.*

Proof. If $\alpha_n = q_n(s^*)$ for every n when all players form a single Market, then $I = \mathcal{N}$ satisfies (i) trivially and (ii) directly, and the construction in the proof of Proposition 5 (Conditions for Social Order), with $R = \emptyset$, has $E^C = \emptyset$, so it is a Trade equilibrium. Conversely, in a Trade equilibrium every component is a Market or an autarkic singleton. Each Market component satisfies (i) and (ii) by the necessity half of Proposition 5, so by Step 3 of that proof its members satisfy (20). Each autarkic player's bundle is proportional to $Y(\mathcal{N})$ by Corollary 1 (Law of One Power Price), so by the equality case of Lemma C.6 (Value Bound) and Theorem 1 (Characterization) her production

value equals her payoff, $q_n U(Y(\mathcal{N}))$. Every player therefore satisfies (20). By (19) with $I = \mathcal{N}$ and (18), $\alpha_n = q_n(s^*)$ for every n.

■

C.2 Proofs for Section 4

Corollary 5 ***Transfer Schedule*****:** *For any strategy profile σ, there exists a transfer schedule of consumer goods, applied to the final allocation at σ, under which every player's payoff equals $q_n\, U(Y(\mathcal{N}))$.*

Proof. Fix σ. In every realization of the outcome lottery $g(h(\sigma))$, each component's output is allocated in full among its members, so the final bundles satisfy $\sum_{n\in\mathcal{N}} X_n^F = Y(\mathcal{N})$. In each realization, transfer $T_n = q_n\, Y(\mathcal{N}) - X_n^F$ to player n. Because the q_n sum to one, $\sum_{n\in\mathcal{N}} T_n = Y(\mathcal{N}) - Y(\mathcal{N}) = 0$, so the schedule only reallocates the final allocation, and every player receives the bundle $q_n\, Y(\mathcal{N})$ with certainty. By homogeneity of degree one, her payoff is $U(q_n Y(\mathcal{N})) = q_n\, U(Y(\mathcal{N}))$. In a single Market of singleton candidates, $X_n^F = \alpha_n Y(\mathcal{N})$, so the schedule consists of the transfers $(q_n - \alpha_n)\, Y(\mathcal{N})$.

■

Lemma 2 ***Coalition Optimization*****:** *Fix a Conflict C and a candidate $B \subseteq V(C)$ whose sharing rule assigns each member a share $\gamma_i > 0$ that does not depend on the labor allocation. Hold the action profile, and with it the interaction graph and the partition into candidates, and every labor allocation outside B fixed, and suppose $S_{-B} \equiv \sum_{j\in V(C)\setminus B} S_j > 0$. A profile of labor allocations $(L_i)_{i\in B}$ is a Nash equilibrium of the induced game among the members of B if and only if it maximizes the candidate's joint payoff $p_B\, U(Y(C))$ over $\prod_{i\in B} \mathcal{L}_i$.*

Proof. For every $i \in B$, $\pi_i = \gamma_i\, p_B\, U(Y(C))$. Each γ_i is a positive constant, so every member's payoff is a positive multiple of the same function, and the induced game is an identical-interest game with common payoff $p_B\, U(Y(C))$. Any maximizer of $p_B\, U(Y(C))$ is therefore a Nash equilibrium, since no member can raise the common payoff by changing her own allocation.

Conversely, let $L_B = (L_i)_{i\in B}$ be a Nash equilibrium of the induced game. The argument is that each member's best response rules out improvement along her own direction,

these combine into a bound along any joint direction, and concavity of the log payoff turns that bound into global optimality. The first steps establish the concavity and finiteness this requires.

Positivity. $p_B\,U\big(Y(C)\big) > 0$ at L_B. Otherwise any member i could deviate to an allocation with $0 < s_i < \tau$ and $t_{im} > 0$ for every m. Then $S_B \geq f_{iS}(s_i) > 0$, so $p_B > 0$, and every component of $Y(C)$ is positive, so $U\big(Y(C)\big) > 0$ because every consumer good is essential.

Concavity. Because $S_{-B} > 0$, p_B is concave in the members' security by Lemma C.1 (Concavity of the Contest Share), and $U\big(Y(C)\big)$ is concave in the members' consumer-good labor by Lemma C.3 (Concavity of Utility in Labor). The set P of coalition profiles at which $p_B\,U\big(Y(C)\big) > 0$ is the intersection of the sets on which $p_B > 0$ and $U\big(Y(C)\big) > 0$; each is convex, because a concave function that is positive at two points is positive on the segment between them, so P is convex. On P, $\ln p_B$ and $\ln U\big(Y(C)\big)$ are concave, as the increasing, concave function ln of positive concave functions, so the log payoff $\ln p_B + \ln U\big(Y(C)\big)$ is concave on P.

Positive marginal values. At L_B, $U\big(Y(C)\big) > 0$, so every component of $Y(C)$ is positive because every consumer good is essential, and $\partial U\big(Y(C)\big)/\partial X_m > 0$ for every m by Lemma C.4 (Positive Marginal Utility). Also $dp_B/dS_B = S_{-B}/(S_B + S_{-B})^2 > 0$. Since both factors are positive, $p_B\,U\big(Y(C)\big)$ is strictly increasing in each member's output of security and of every consumer good.

Finite marginal products. Each production function is C^2 at positive labor, so its derivative is finite wherever a member allocates positive labor; at zero labor, its right derivative may be infinite. At L_B this cannot occur for any member. Suppose member i allocates zero labor to one of her activities, with production function f and $f'(0^+) = \infty$. Since $\tau > 0$, she allocates positive labor ℓ_j to some other activity j, with production function f_j. Let her move $\varepsilon \in (0, \ell_j/2)$ units of labor from j to the first activity. Output of the first activity rises by $f(\varepsilon)$, where $f(\varepsilon)/\varepsilon \to \infty$ as $\varepsilon \to 0$, while by concavity output of j falls by at most $f_j'(\ell_j/2)\,\varepsilon$, a finite multiple of ε. Because $p_B\,U\big(Y(C)\big)$ is differentiable in the members' aggregate outputs at L_B and, by the preceding paragraph, strictly increasing in each, its change divided by ε tends to $+\infty$. Hence the move raises her payoff $\pi_i = \gamma_i\,p_B\,U\big(Y(C)\big)$ for every sufficiently small $\varepsilon > 0$: a profitable unilateral deviation. For every member and every activity, the marginal product at L_B, using right derivatives where labor is zero, is therefore finite.

Global optimality. Take any coalition profile L_B'. If $p_B\,U\big(Y(C)\big) = 0$ at L_B', then L_B' is worse than L_B. Otherwise $L_B' \in P$. Let $d_i = L_i' - L_i$ and $d = (d_i)_{i \in B}$; each d_i is a feasible direction on member i's own simplex, because each member's labor constraint involves only her own labor. For $\lambda \in [0, 1]$, let $h(\lambda)$ be the log payoff $\ln p_B + \ln U\big(Y(C)\big)$ at $L_B + \lambda d$, and $h_i(\lambda)$ the log payoff at $\big(L_i + \lambda d_i,\, L_{B \setminus \{i\}}\big)$. Both paths remain in P because P is convex. Since L_i is a best response and ln is increasing, $h_i(\lambda) \leq h_i(0)$ for every λ, so the right derivative $h_i'(0^+)$ is at most zero, and it is finite by the preceding paragraph. Because the log payoff depends on the members' labor only through the sums S_B and $Y(C)$, and is differentiable in those sums at L_B, the chain rule gives $h'(0^+) = \sum_{i \in B} h_i'(0^+) \leq 0$. Finally, h is concave because the log payoff is concave on P, so its chord slopes from 0 are nonincreasing, and $h(1) - h(0) \leq h'(0^+) \leq 0$. Since ln is increasing, $p_B\,U\big(Y(C)\big)$ is no larger at L_B' than at L_B. Hence L_B maximizes $p_B\,U\big(Y(C)\big)$.

■

Proposition 6 ***Shares Must Match Power*****:** *Suppose a coalition B, with $|B| \geq 2$, forms on the path of a strategy profile σ. If $\gamma_i \neq S_i/S_B$ for any $i \in B$, then σ is not a subgame perfect equilibrium.*

Proof. Suppose σ is an SPE, and let H be the component containing B. By Proposition 4 (Security Budget), every player holds positive security, so $S_H > 0$. By Theorem 1 (Characterization) and Lemma 1 (Aggregate Payoff Identity), $U\big(Y(H)\big) = \sum_{j \in V(H)} q_j\, U\big(Y(\mathcal{N})\big) > 0$. I show $\gamma_i = S_i/S_B$ for every $i \in B$.

Floors. Any player $j \in V(H)$ can declare theft against every other member of H. Since H is a connected component, the resulting Conflict has vertex set exactly $V(H)$ and prize $Y(H)$, and j is a singleton in it. By (13) her payoff is $\frac{S_j}{S_H} U\big(Y(H)\big)$, whatever the other players' postures. Since σ is an SPE, this deviation is not profitable:

$$\pi_j(\sigma) \geq \frac{S_j}{S_H} U\big(Y(H)\big) \qquad \forall\, j \in V(H).$$

Squeeze. By Lemma 1, $\sum_{j \in V(H)} \pi_j(\sigma) = U\big(Y(H)\big)$, which is also the sum of the floors, since $\sum_{j \in V(H)} S_j = S_H$. Every floor therefore binds: $\pi_j(\sigma) = \frac{S_j}{S_H} U\big(Y(H)\big)$ for every $j \in V(H)$.

Positive security. Suppose $S_j = 0$ for some $j \in V(H)$, so that $\pi_j(\sigma) = 0$. Because $U\big(Y(H)\big) > 0$, every component of $Y(H)$, and hence of $Y(\mathcal{N})$, is positive, so j can shift a small amount of labor into security while keeping $U\big(Y(\mathcal{N})\big)$ positive, by continuity, and then declare theft against every other player. Whatever the others do, the resulting Conflict spans $\mathcal{N}$, her security is positive, and her payoff $q_j\, U\big(Y(\mathcal{N})\big)$ is positive: a profitable deviation. Hence $S_j > 0$ for every $j \in V(H)$, and in particular $S_B > 0$.

Shares. Each member of B receives $\pi_i(\sigma) = \gamma_i W_B$, where W_B is the candidate's payoff: $p_B\, U\big(Y(H)\big)$ if H is a Conflict and $\alpha_B\, U\big(Y(H)\big)$ if H is a Market. Summing the binding floors over B gives $W_B = \frac{S_B}{S_H} U\big(Y(H)\big) > 0$, and dividing,

$$\gamma_i = \frac{\pi_i(\sigma)}{W_B} = \frac{S_i}{S_B} \qquad \forall\, i \in B.$$

Under equal shares, $\gamma_i = 1/|B|$, so $S_i = S_B/|B|$ for every member.

■

Proposition 7 ***Grand Coalition Instability*****:** *If $\mathcal{N}$ is a coalition at a strategy profile σ, then σ is not a subgame perfect equilibrium.*

Proof. Suppose $\mathcal{N}$ is a coalition at σ.

The coalition disarms. Every player is a member, so the members' total payoff is $\sum_H U(Y(H)) \leq U(Y(\mathcal{N}))$ by Lemma 1 (Aggregate Payoff Identity) and Lemma C.5 (Superadditivity of U), and the members attain $U(Y(\mathcal{N}))$ at any labor profile by all declaring trade toward one another, forming a single Market. The coalition's joint problem is therefore to maximize $U(Y(\mathcal{N}))$ over labor. Its maximum is positive, since a single player can allocate positive labor to every good, so at the optimum every component of $Y(\mathcal{N})$ is positive and, by Lemma C.4 (Positive Marginal Utility), U is strictly increasing in each good. Security does not enter $Y(\mathcal{N})$, and each production function is strictly increasing, so any player with $s_n > 0$ could raise $U(Y(\mathcal{N}))$ by moving that labor into a consumer good. Hence $s_n = 0$ for every $n \in \mathcal{N}$, and the members' payoffs sum to $U(Y(\mathcal{N})) > 0$.

Conquest. Since payoffs sum to $U(Y(\mathcal{N}))$ and $N \geq 2$, some player j receives $\pi_j(\sigma) < U(Y(\mathcal{N}))$. Let j shift $\varepsilon > 0$ units of labor from a consumer good she produces into security and declare theft against every other player. Whatever the others' postures, the resulting Conflict spans $\mathcal{N}$. Every other player still holds no security while j holds $f_{jS}(\varepsilon) > 0$, so by (13) j wins with certainty and obtains $U(Y^{\varepsilon}(\mathcal{N}))$, where $Y^{\varepsilon}(\mathcal{N})$ is aggregate production after her shift. By continuity of production and utility, $U(Y^{\varepsilon}(\mathcal{N})) \to U(Y(\mathcal{N})) > \pi_j(\sigma)$ as $\varepsilon \to 0$, so the deviation is profitable for small ε. Hence σ is not a subgame perfect equilibrium.

■

Proposition 8 ***No Productive Coalitions*****:** *In any SPE, every member of every coalition receives the payoff, and chooses the labor allocation, she would as a lone combatant. Consequently, no SPE contains a coalition consisting of two or more members, with at least one member who produces a consumer good.*

Proof. Let σ be an SPE. By Theorem 1 (Characterization), its labor profile is a Nash equilibrium of the universal-Conflict labor-allocation game, and every player's payoff is $\frac{S_j}{S_{\mathcal{N}}} U(Y(\mathcal{N}))$, where $S_{\mathcal{N}} = \sum_{j \in \mathcal{N}} S_j$. By Proposition 4 (Security Budget) and its proof, every player holds positive security and $U(Y(\mathcal{N})) > 0$, so every component of $Y(\mathcal{N})$ is positive and U is differentiable at $Y(\mathcal{N})$.

Lone-combatant payoffs and choices. Each player's equilibrium payoff is $\frac{S_i}{S_{\mathcal{N}}} U(Y(\mathcal{N}))$, her payoff as a singleton in the universal Conflict. Moreover, any member i of any coalition can deviate to any labor allocation and declare theft against every other player. Whatever the others do, the resulting Conflict spans $\mathcal{N}$ and i is a singleton in it, so she obtains $\frac{S_i}{S_{\mathcal{N}}} U(Y(\mathcal{N}))$ evaluated at her new allocation. Since σ is an SPE, no such deviation is profitable, so her equilibrium allocation maximizes $\frac{S_i}{S_{\mathcal{N}}} U(Y(\mathcal{N}))$ given the other players' labor: she chooses exactly as a lone combatant would.

No productive coalitions. Suppose B is a coalition with $|B| \geq 2$ and some member $i \in B$ allocates $t_{im} > 0$ to good m. Every member holds positive security, so $0 < S_i < S_B$.

Since B is a coalition, its members' labor allocations and postures jointly maximize their total payoff, given the other players' strategies. Among the joint plans available to them is any labor profile combined with a posture choice that does not depend on what the others do. If $B \neq \mathcal{N}$, the members can declare theft as a bloc against every player outside B; the resulting Conflict spans $\mathcal{N}$, no Conflict edge joins two members, and B wins with probability $S_B/S_{\mathcal{N}}$, so the members obtain $\frac{S_B}{S_{\mathcal{N}}}U\big(Y(\mathcal{N})\big)$ in total. If $B = \mathcal{N}$, the members can all declare trade toward one another, forming a single Market whose payoffs sum to $U\big(Y(\mathcal{N})\big) = \frac{S_B}{S_{\mathcal{N}}}U\big(Y(\mathcal{N})\big)$ by Lemma 1 (Aggregate Payoff Identity). In either case the members can secure $\frac{S_B}{S_{\mathcal{N}}}U\big(Y(\mathcal{N})\big)$ at any labor profile of their choosing. Their equilibrium total equals this expression at the equilibrium profile, so their equilibrium labor profile maximizes it.

Since $s_i > 0$ and $t_{im} > 0$, the relevant derivatives are finite. Because member i can shift labor between security and good m in either direction, individual optimality requires

$$f'_{iS}(s_i)\Big(\frac{1}{S_i} - \frac{1}{S_{\mathcal{N}}}\Big) = \frac{\partial \ln U\big(Y(\mathcal{N})\big)}{\partial t_{im}},$$

while joint optimality requires that moving i's labor from security to good m not raise $\ln\big(\frac{S_B}{S_{\mathcal{N}}}U(Y(\mathcal{N}))\big)$:

$$f'_{iS}(s_i)\Big(\frac{1}{S_B} - \frac{1}{S_{\mathcal{N}}}\Big) \geq \frac{\partial \ln U\big(Y(\mathcal{N})\big)}{\partial t_{im}}.$$

Since $f'_{iS}(s_i) > 0$, together these give $\frac{1}{S_B} \geq \frac{1}{S_i}$, contradicting $S_i < S_B$. Hence no coalition with two or more members contains a member who produces a consumer good.

Finally, if every consumer-good production function has an infinite right derivative at zero, the proof of Proposition 4 shows that every player produces every good, so every coalition with two or more members is productive, and none exists in any SPE.

■

D Corner Solutions in Optimal Labor Allocations

In the optimality conditions of Proposition 4, security is always strictly positive. Corners in consumer goods are innocuous: a player who produces none of some good satisfies the corresponding condition as an inequality, and aggregate production of every good remains positive because every good is essential. When a good's production function has an infinite right derivative at zero, every player produces it.

The corner $s_n = \tau$, complete specialization in security, deserves comment, because it describes a player's position in the economy rather than a technicality. It can occur only if every consumer-good

production function has a finite right derivative at zero, and it requires

$$(1 - q_n)\,\frac{f'_{nS}(\tau)}{f_{nS}(\tau)} \;\geq\; \max_{m \in \mathcal{M}} \frac{\partial U(Y(\mathcal{N}))/\partial X_m}{U(Y(\mathcal{N}))}\, f'_{nm}(0),$$

with q_n evaluated at $s_n = \tau$. The left side is the proportional return to security; each term on the right is the proportional return to beginning production of a consumer good. Two archetypes satisfy this condition from opposite ends of the distribution of power.

The *predator* dominates the economy with q_n at or near the maximum attainable value, so the left side is small, since additional security barely raises a share that is already nearly complete. Complete specialization then requires that her productivity in consumer goods be so low relative to her productivity in security that even this small return dominates. The predator specializes not because conflict is especially rewarding at the margin, but because producing consumer goods is all but useless to her.

The *desperado* is nearly certain to lose: q_n is close to zero, so the left side is approximately the proportional return to her own security production. The two sides of the condition are measured against different bases. The return to security is proportional to her own security output, which is small, while the return to a consumer good is proportional to the economy's total output, to which she contributes little. A player with little to contribute to the economy can therefore find that security offers the larger proportional return, despite long odds of winning.

Security specialists are also the only players the model allows to form coalitions in equilibrium. By Proposition 8, a coalition with two or more members can exist in an SPE only if none of its members produces a consumer good. A coalition of players who each hold only security can satisfy every equilibrium requirement: its shares must match its members' power (Proposition 6), each member's corner must be her own best response, and the corner must also maximize the coalition's joint payoff. The last requirement is the most demanding. Shifting member i's labor into good m must not raise the coalition's total,

$$f'_{iS}(\tau)\Big(\frac{1}{S_B} - \frac{1}{S_{\mathcal{N}}}\Big) \;\geq\; \frac{\partial U(Y(\mathcal{N}))/\partial X_m}{U(Y(\mathcal{N}))}\, f'_{im}(0),$$

a stricter condition than her individual corner condition, because $S_B > S_i$. Members of such

a coalition must be even less productive than a lone security specialist: enough that not even the coalition's collective interest would put any of them to productive work. Yet the coalition accomplishes nothing. Each member receives her lone-combatant payoff and chooses exactly what she would choose alone. The one coalition the model sustains is a gang whose joint optimization is an empty promise.

By Theorem 1, both archetypes receive $q_n\, U\big(Y(\mathcal{N})\big)$ in equilibrium, whether working alone or in a gang. Neither produces anything a Market would pay for, so both must fight for their payoff. The predator expects to win the prize that others produce; the desperado, with no better option, takes a long shot at it. Both live entirely on the production of others, one through dominance and the other through desperation.